



MiniPoly: Automatic Extraction of Efficient JavaScript Polyfill from Language Specification

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- Research Field - **Programming Languages and Software Engineering**
 - Program Analysis
 - Mechanized Language Specification
 - Automated Testing
 - Program Synthesis & Repair
- **Recent Publications (2020~)**
 - **PLDI** (2023, 2024) / **SAS** (2026)
 - **ICSE** (2021) /
FSE (2021, 2022, 2025-Demo) /
ASE (2020, 2021, 2025, 2026)
 - **CACM** (2024) / **CSUR** (2021) /
TOSEM (2026)



Built-in APIs in JavaScript

ES2025

Native Code
(e.g., C++, Java)

```
const a = new Set([1, 2, 3]);  
const b = new Set([1, 2]);  
a.isSupersetOf(b); // true
```



≤ 126



≤ 2025.09



≥ 122



≤ 16.7



≤ 21



Polyfill

- **Polyfill** is a **JavaScript** implementation of built-in APIs
- **core-js** is used by **41% of websites** and **$\geq 46\text{M}$ downloads per week**

```
module.exports = function isSupersetOf(other) {  
  var 0 = aSet(this);  
  var otherRec = getSetRecord(other);  
  if (size(0) < otherRec.size) return false;  
  var iterator = otherRec.getIterator();  
  return iterateSimple(iterator, function (e) {  
    if (!has(0, e)) {  
      return iteratorClose(iterator, 'normal', false);  
    }  
  }) !== false;  
};
```

core-js

Polyfill

```
import 'core-js/actual/set/is-superset-of';
const a = new Set([1, 2, 3]);
const b = new Set([1, 2]);
a.isSupersetOf(b); // true
```

```
module.exports = function isSupersetOf(other) {
  var 0 = aSet(this);
  var otherRec = getSetRecord(other);
  if (size(0) < otherRec.size) return false;
  var iterator = otherRec.getIterator();
  return iterateSimple(iterator, function (e) {
    if (!has(0, e)) {
      return iteratorClose(iterator, 'normal', false);
    }
  }) !== false;
};
```

core-js



≤ 126



≤ 2025.09



≥ 122



≤ 16.7



≤ 21



ECMA-262 - JavaScript Language Specification

Built-in

Set.prototype.isSupersetOf (*other*)

1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < *otherRec*.[[Size]], return **false**.
5. Let *keyslter* be ? GetIteratorFromMethod(*otherRec*.[[SetObject]], *otherRec*.[[Keys]]).
6. Let *next* be NOT-STARTED.
7. Repeat, while *next* is not DONE,
 - a. Set next to ? IteratorStepValue(*keyslter*).
 - b. If *next* is not DONE, then
 - i. If SetDataHas(*O*.[[SetData]], *next*) is **false**, then
 1. Perform ? IteratorClose(*keyslter*, NormalCompletion(UNUSED)).
 2. Return **false**.
8. Return **true**.

tc39/ecma262

Status, process, and documents for ECMA-262



Syntax

*AdditiveExpression*_[Yield, Await] :

*MultiplicativeExpression*_[?Yield, ?Await]

*AdditiveExpression*_[?Yield, ?Await] + *MultiplicativeExpression*_[?Yield, ?Await]

*AdditiveExpression*_[?Yield, ?Await] - *MultiplicativeExpression*_[?Yield, ?Await]

AdditiveExpression : *AdditiveExpression* + *MultiplicativeExpression*

1. Return ? EvaluateStringOrNumericBinaryExpression(
AdditiveExpression, +, *MultiplicativeExpression*).

Semantics

Current Approach - Manual Translation

Built-in

Set.prototype.isSupersetOf (*other*)

1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < *otherRec*.[[Size]], return **false**.
5. Let *keysIter* be ? GetIteratorFromMethod(*otherRec*.[[SetObject]], *otherRec*.[[Keys]]).
6. Let *next* be NOT-STARTED.
7. Repeat, while *next* is not DONE,
 - a. Set next to ? IteratorStepValue(*keysIter*).
 - b. If *next* is not DONE, then
 - i. If SetDataHas(*O*.[[SetData]], *next*) is **false**, then
 1. Perform ? IteratorClose(*keysIter*, NormalCompletion(UNUSED)).
 2. Return **false**.
8. Return **true**.

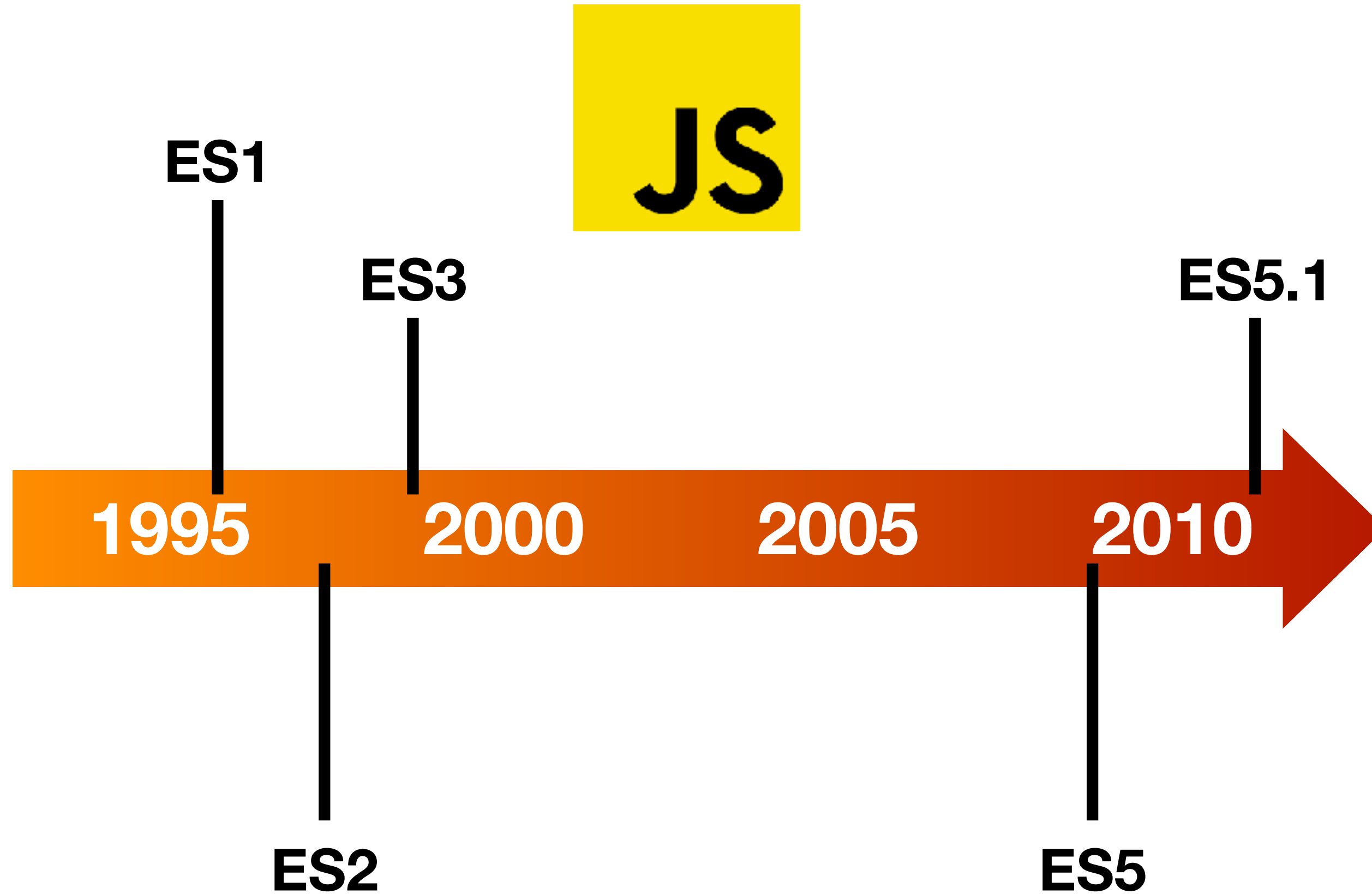
Manual

TC
39

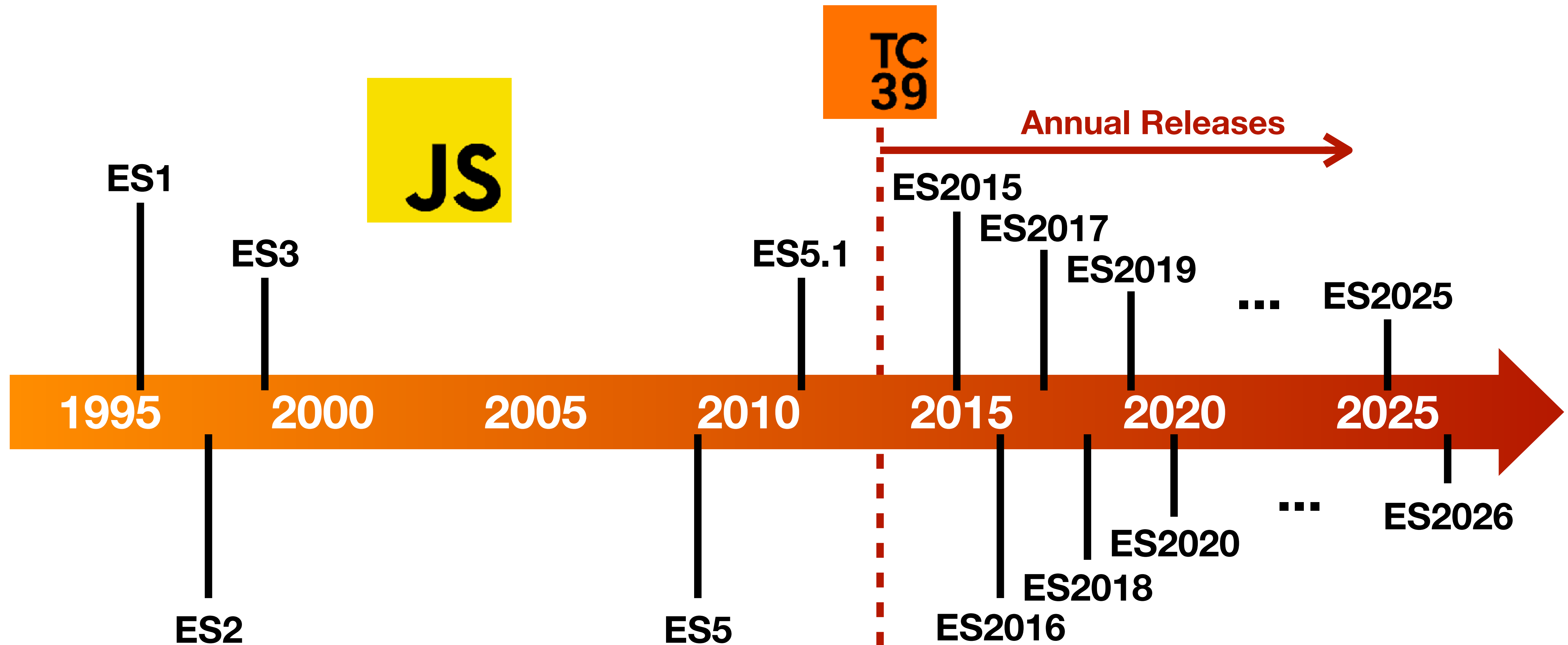
```
function isSupersetOf(other) {  
  var O = aSet(this);  
  var otherRec = getSetRecord(other);  
  if (size(O) < otherRec.size) return false;  
  var iterator = otherRec.getIterator();  
  return iterateSimple(iterator,  
    function (e) {  
      if (!has(O, e)) {  
        return iteratorClose(iterator,  
          'normal', false);  
      }  
    }) !== false;  
};
```

core-js

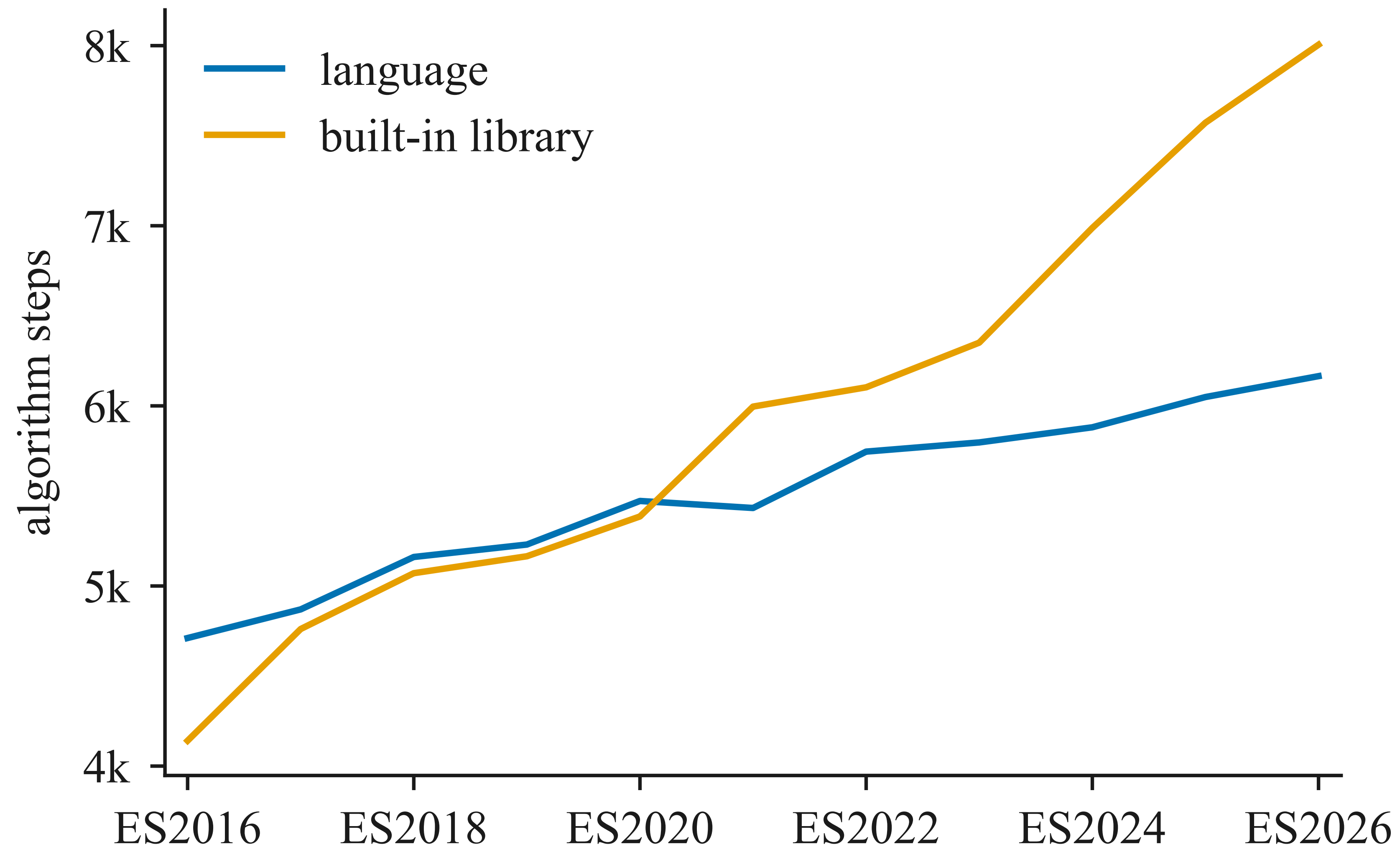
Problem - **Fast Evolving** JavaScript



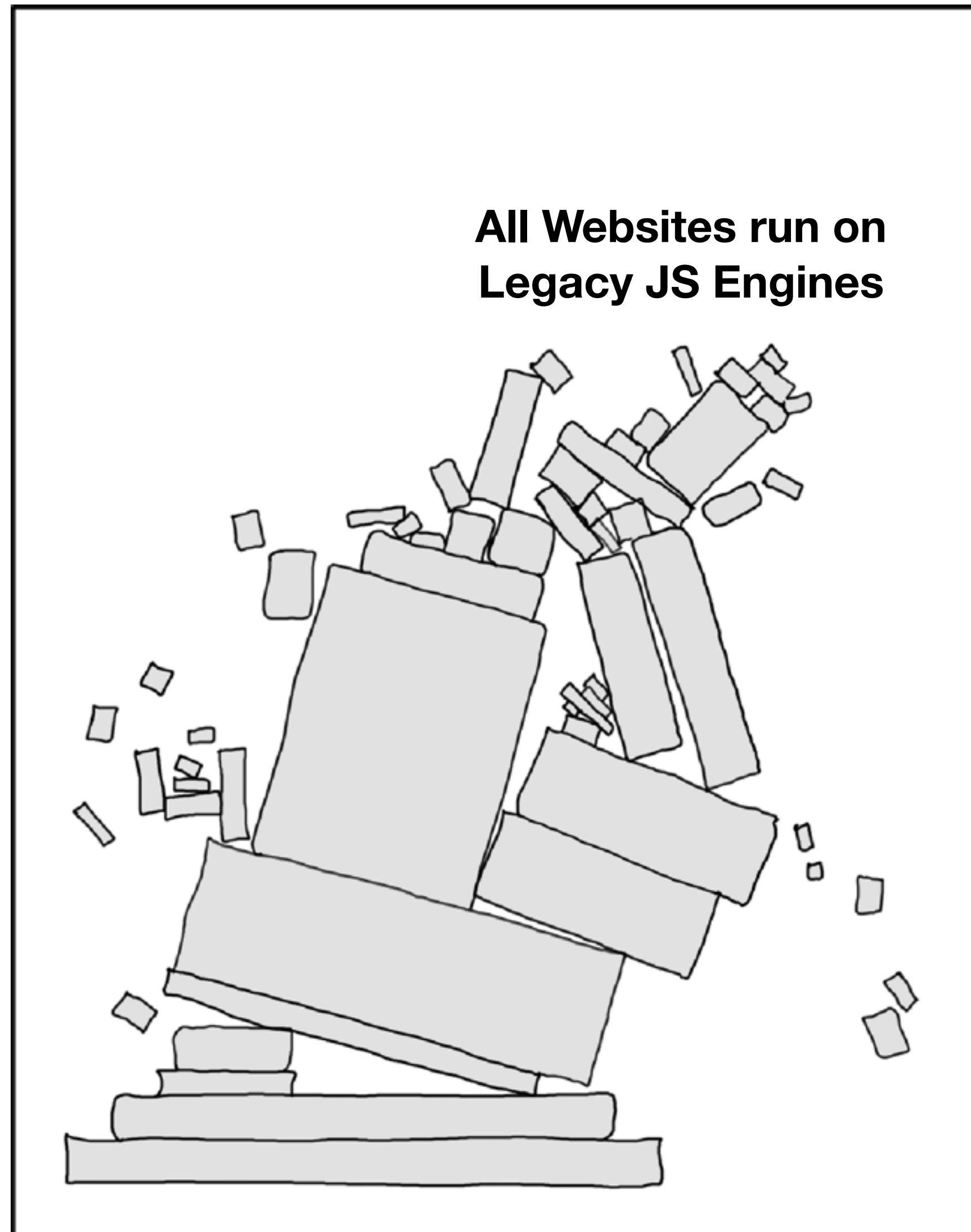
Problem - **Fast Evolving** JavaScript



Problem - **Fast Evolving** JavaScript



Problem - Burden of **Manual** Implementation



core-js

Our Approach - Automatic Extraction from ECMA-262

Built-in

Set.prototype.isSupersetOf (*other*)

1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < *otherRec*.[[Size]],
return **false**.
5. Let *keyslter* be ? GetIteratorFromMethod(
otherRec.[[SetObject]], *otherRec*.[[Keys]]).
6. Let *next* be NOT-STARTED.
7. Repeat, while *next* is not DONE,
 - a. Set next to ? IteratorStepValue(*keyslter*).
 - b. If *next* is not DONE, then
 - i. If SetDataHas(*O*.[[SetData]], *next*) is **false**, then
 1. Perform ? IteratorClose(*keyslter*,
NormalCompletion(UNUSED)).
 2. Return **false**.
8. Return **true**.

Extract

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Polyfill

in **JavaScript**

Challenges - **Correctness** and **Efficiency**

Built-in

Set.prototype.isSupersetOf (*other*)

1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O*, [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O*.[[SetData]]) < *otherRec*.[[Size]],
return **false**.
5. Let *keyslter* be ? GetIteratorFromMethod(
otherRec.[[SetObject]], *otherRec*.[[Keys]]).
6. Let *next* be NOT-STARTED.
7. Repeat, while *next* is not DONE,
 - a. Set next to ? IteratorStepValue(*keyslter*).
 - b. If *next* is not DONE, then
 - i. If **SetDataHas**(*O*.[[SetData]], *next*) is **false**, then
 1. Perform ? IteratorClose(*keyslter*,
NormalCompletion(UNUSED)).
 2. Return **false**.
8. Return **true**.

1) How to **correctly** handle *internal repr. of control flows*?

2) How to extract *algorithmically efficient* polyfill?

Extract

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Polyfill

in **JavaScript**

Challenge 1 - **Correctness** (Internal Repr. of Control Flow)

A computation **X** $\left\{ \begin{array}{ll} \text{is **normally** terminated with a **value**} & \text{if } X \text{ is a normal completion} \\ \text{throws an **exception**} & \text{if } X \text{ is a throw completion} \end{array} \right.$

IteratorClose (*iteratorRecord*, *completion*)

...

3. Let *innerResult* be **GetMethod**(...).

4. If *innerResult* is a **normal completion**, then

a. Let *return* be *innerResult*.[[Value]].

b. If *return* is undefined, return ? *completion*.

c. Set *innerResult* to **Call**(...).

5. If *completion* is a **throw completion**, return ? *completion*.

6. If *innerResult* is a **throw completion**, return ? *innerResult*.

...

8. Return ? *completion*.

return *completion*
OR

throw *completion*

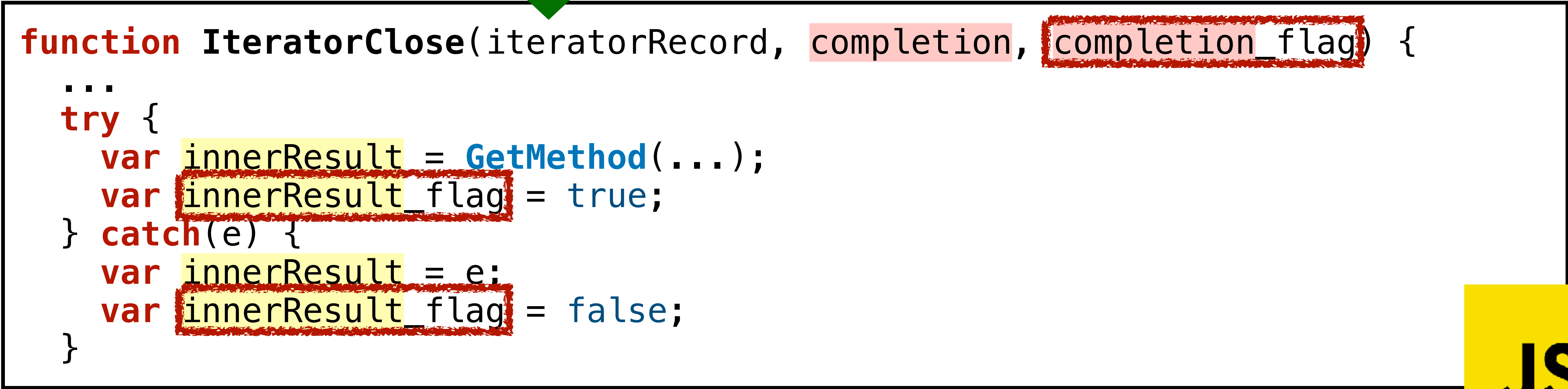
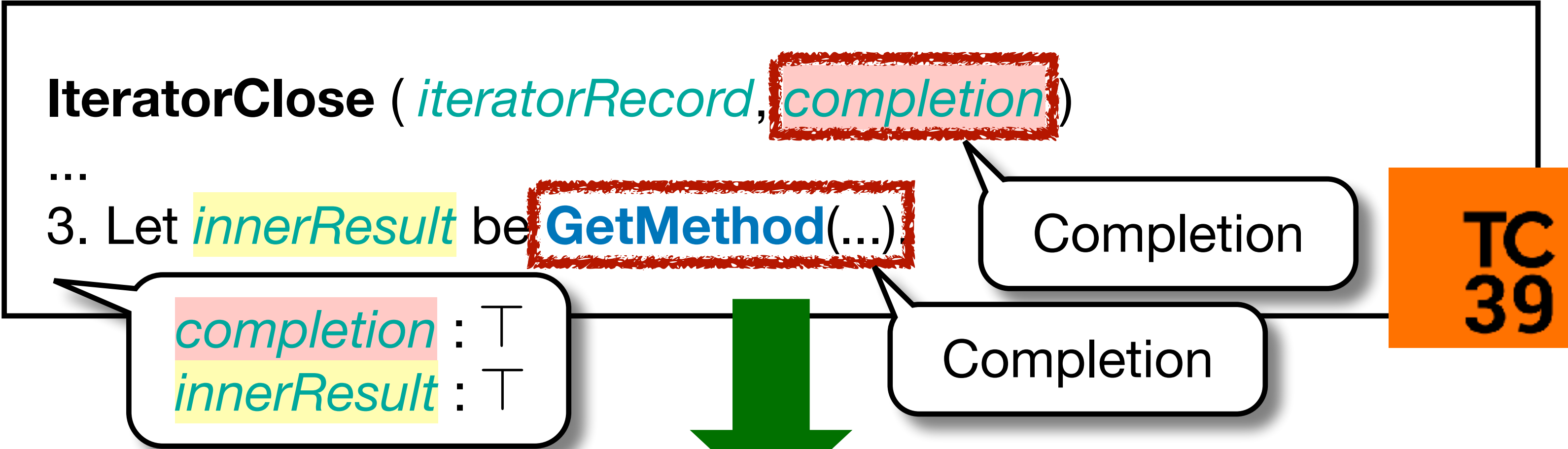
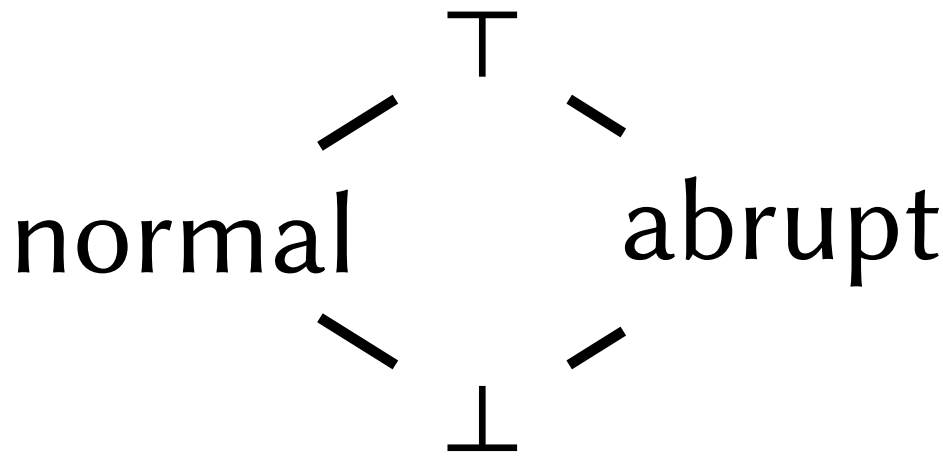
throw *completion*

throw *innerResult*

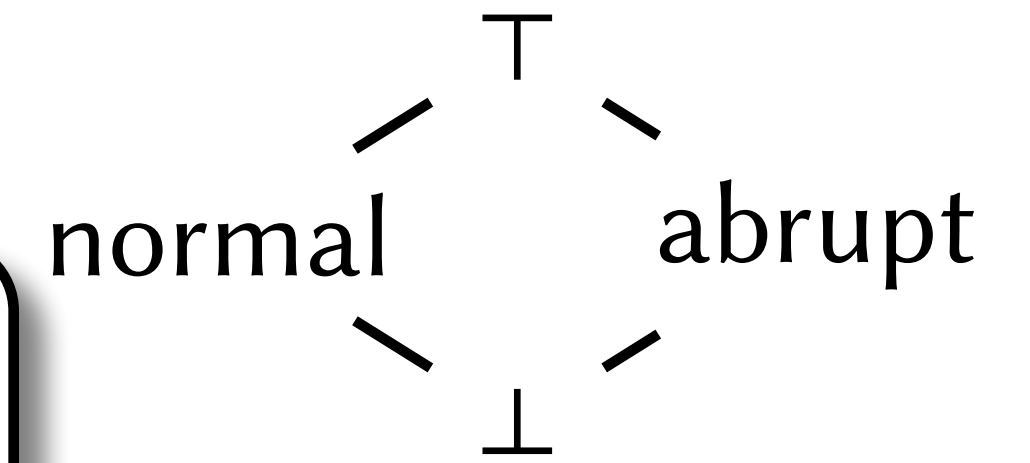
return *completion*

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Solution 1 - Completion Type Analysis



Solution 1 - Completion Type Analysis



4. If *innerResult* is a **normal completion** then
- Let *return* be *innerResult*.[[Value]].
 - If *return* is undefined, return ? *completion*
 - Set *innerResult* to **Call**(...).

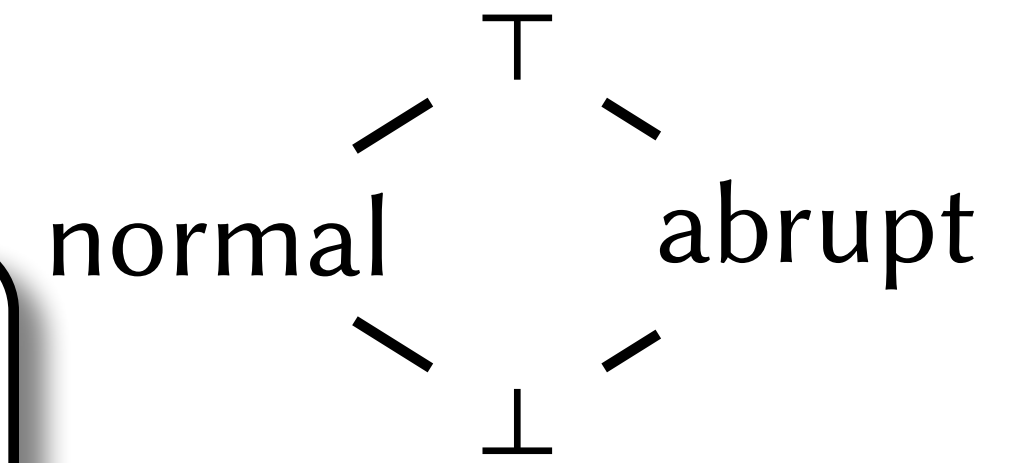
completion : T
innerResult : normal

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```
if (innerResult_flag) {  
  ...  
}
```

JS

Solution 1 - Completion Type Analysis



4. If **innerResult** is a **normal completion** then
- Let **return** be **innerResult**.[[Value]].
 - If **return** is undefined, return ? **completion**
 - Set **innerResult** to **Call(...)**.

completion : T
innerResult : normal

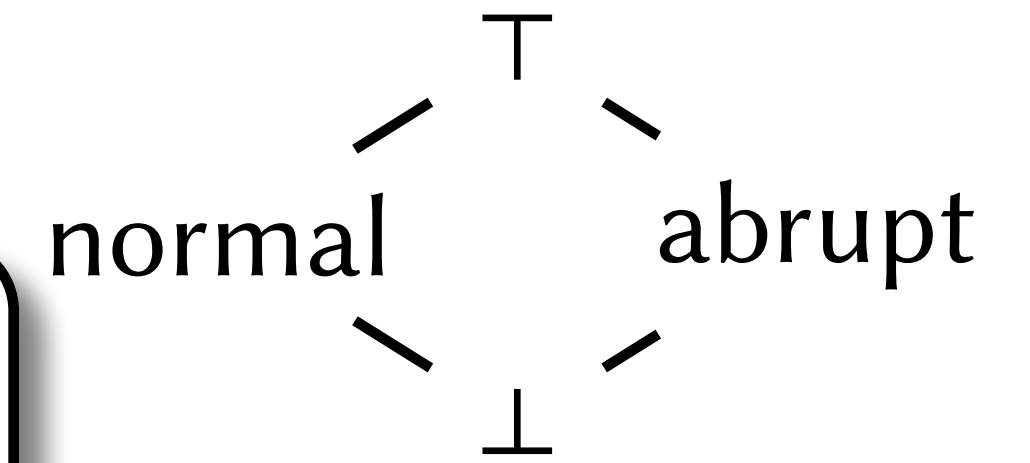
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Completion

```
if (innerResult_flag) {  
  ...  
  if ( return === undefined) {  
    if (completion_flag) return completion;  
    else throw completion;  
  }  
}
```

JS

Solution 1 - Completion Type Analysis



4. If **innerResult** is a **normal completion** then
- Let **return** be **innerResult**.[[Value]].
 - If **return** is undefined, return ? **completion**
 - Set **innerResult** to **Call(...)**.

completion : T
innerResult : normal

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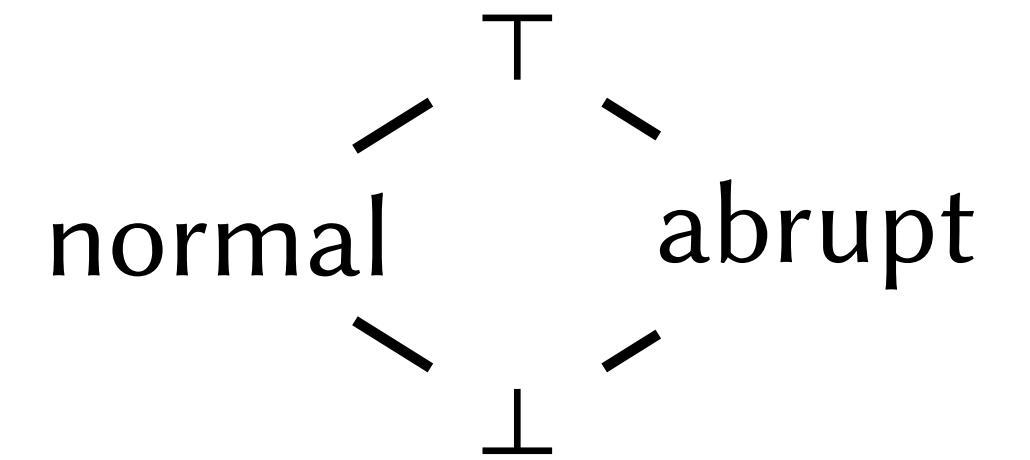
innerResult : T

Completion

```
if (innerResult_flag) {  
  ...  
  if ( return === undefined) {  
    if (completion_flag) return completion;  
    else throw completion;  
  }  
  try { innerResult = Call(...); innerResult_flag = true; } catch ...  
}
```

JS

Solution 1 - Completion Type Analysis



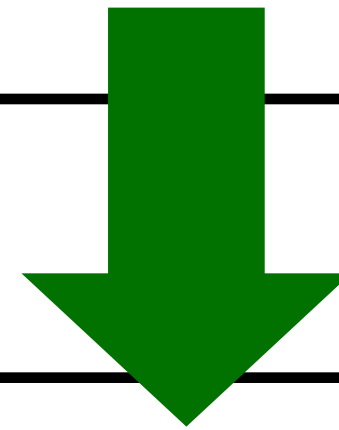
5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
- ...
8. Return ? *completion*.

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completion : abrupt

innerResult : abrupt

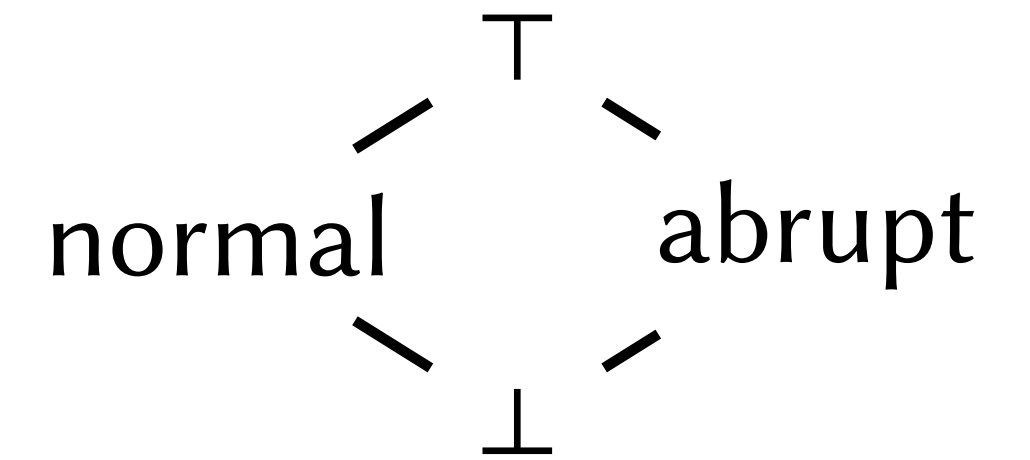
completion : normal



```
if (!completion_flag) throw completion;  
if (!innerResult_flag) throw innerResult;
```

JS

Solution 1 - Completion Type Analysis



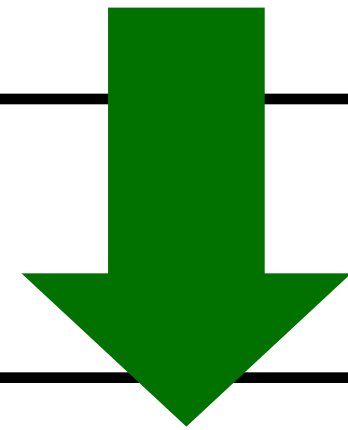
5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
- ...
8. Return ? *completion*.

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completion : abrupt

innerResult : abrupt

completion : normal



```
if (!completion_flag) throw completion;  
if (!innerResult_flag) throw innerResult;  
...  
return completion;
```

JS

Solution 1 - Completion Type Analysis

```
function IteratorClose(iteratorRecord, completion, completion_flag) {  
    ...  
    try { var innerResult = GetMethod(...); var innerResult_flag = true; }  
    catch(e) { var innerResult = e; var innerResult_flag = false; }  
    if (innerResult_flag) {  
        ...  
        if (_return === undefined) {  
            if (completion_flag) return completion;  
            else throw completion;  
        }  
        try { innerResult = Call(...); innerResult_flag = true; } catch ...  
    }  
    if (!completion_flag) throw completion;  
    if (!innerResult_flag) throw innerResult;  
    ...  
    return completion;  
}
```

JS

Challenge 2- **Efficiency**

$$\mathcal{O}(n)$$

vs.

$$\mathcal{O}(1)$$

Set.prototype.add (*value*) **Linear search**

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...
4. For each element *e* of *S*.[[SetData]], do
a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

Set.prototype. delete (*value*)

...
4. For each element *e* of *S*.[[SetData]], do
a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

Set.prototype. has (*value*)

...
4. For each element *e* of *S*.[[SetData]], do
a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then
...

core-js

Set.prototype.add

Set.prototype.delete

Set.prototype.has

Solution 2 - Rewrite Rules

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not EMPTY and SameValue({{ elem }},
      {{ value : var }}) is *true*, then
      1. {{ body : step }}
```

Set.prototype.add (*value*)

...

4. For each element *e* of `S.[[SetData]]`, do
a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

...

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Solution 2 - Rewrite Rules

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not EMPTY and SameValue({{ elem }},
      {{ value : var }}) is *true*, then
      1. {{ body : step }}
where:
- isSetData(base)
- { isSetData: "_.[[SetData]]" }
```

Set.prototype.add (*value*)

...

4. For each element *e* of **S.[[SetData]]** do
a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

...

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39

Solution 2 - Rewrite Rules

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not ~empty~ and SameValue({{ elem }},
      {{ value : var }}) is *true*, then
      1. {{ body : step }}
where:
  - isSetData(base)
  - { isSetData: " _.[[SetData]] " }
replace: |
  1. If SET_DATA_HAS({{ base }}, {{ value }}) is *true*, then
    1. {{ body }}
```

Set.prototype.add (*value*)

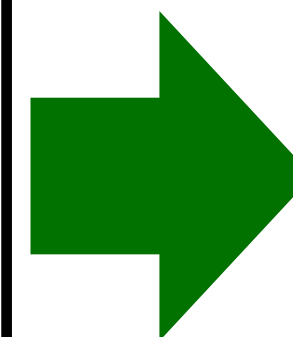
...

4. For each element *e* of *S*.[[SetData]], do

a. If *e* is not EMPTY and SameValue(*e*, *value*) is **true**, then

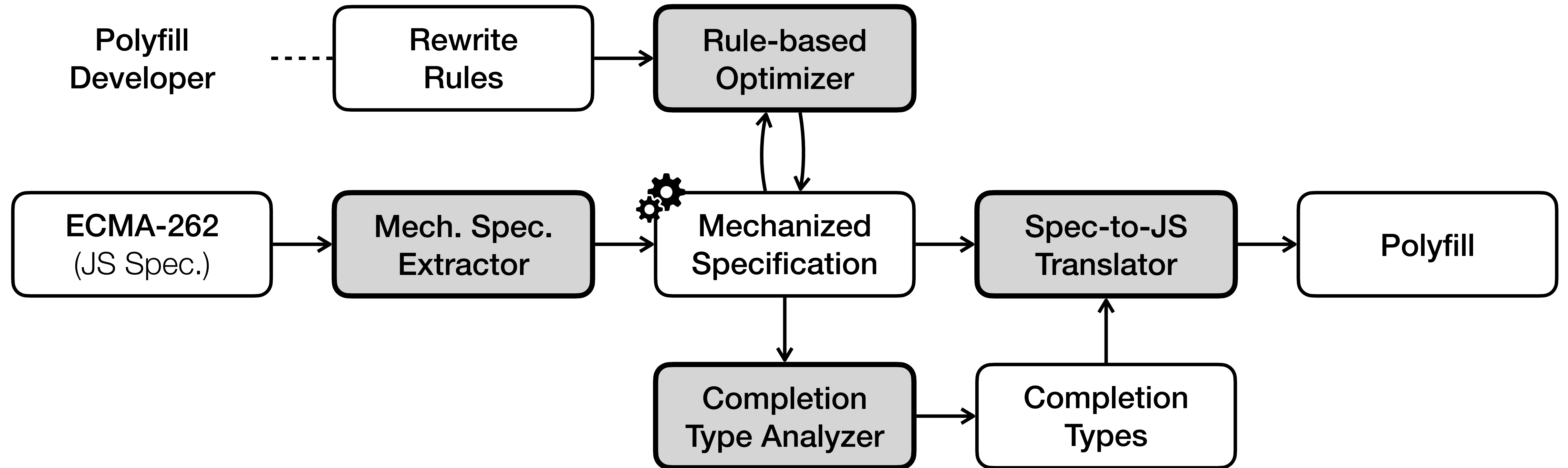
...

TC
39



```
function Set.prototype.add(value) {
  ...
  if (SET_DATA_HAS(S.SetData, value)
      === true) {
    ...
  }
}
```


MiniPoly - Overall Structure



Evaluation

- Target **89 built-in APIs** in **ES2025**
- **Baselines** - `core-js` and `es-shims` (final release versions of 2025)
- Research Questions

RQ1) Correctness

RQ2) Reduced Manual Effort

RQ3) Speed

RQ4) Forward Compatibility

RQ5) Real-World Usability

RQ1) Correctness

Inherently impossible test cases passed by Polyfill implementation

- Tested by **Test262** - official conformance tests
- (# of Fail) **core-js - 172** / **es-shims - 1,058** / **MiniPoly - 16**

Category	# Built-ins	Tests	core-js		es-shims [†]		MiniPoly		Minipoly-opt	
			Pass (%)	Time	Pass (%)	Time	Pass (%)	Time	Pass (%)	Time
Array	24	4,713	4,617 (98.0)	54.3s	4,365 (95.1)	81.5s	4,713 (100.0)	49.2s	4,713 (100.0)	51.5s
String	12	638	616 (96.6)	11.0s	612 (95.9)	12.4s	638 (100.0)	9.7s	638 (100.0)	9.5s
Set	18	748	710 (94.9)	16.1s	698 (93.8)	17.3s	746 (99.7)	13.8s	746 (99.7)	14.4s
Map	12	322	298 (92.5)	8.8s	258 (80.1)	8.4s	320 (99.4)	7.5s	320 (99.4)	7.7s
WeakMap	5	198	168 (84.8)	3.9s	–	–	196 (99.0)	3.6s	196 (99.0)	3.5s
WeakSet	5	166	142 (85.5)	3.9s	–	–	164 (98.8)	3.8s	164 (98.8)	3.1s
Promise	13	1,264	1,154 (91.3)	21.6s	0 (0.0)	10.9s	1,256 (99.4)	18.5s	1,256 (99.4)	18.9s
Total	89	8,049	7,705 (95.7)	119.5s	5,933 (87.4)	130.5s	8,033 (99.8)	106.1s	8,033 (99.8)	108.7s

RQ2) Reduced Manual Effort

Object	#	Baseline		MiniPoly		MiniPoly-opt		Rules
		core-js	es-shims	Auto	Man.	Auto	Man.	
Array	24	2,298	8,978	1,332	317			
String	12	1,281	6,885	418	130			
Set	18	2,313	5,880	864	190	861	283	114 (11)
WeakSet	5	1,882	–	387	123	379	214	
Map	12	1,986	3,062	756	217	747	327	81 (6)
WeakMap	5	2,034	–	446	115	436	231	
Promise	13	2,518	8,031	1,511	203			
Total†	89	14,312	32,836	5,714	1,295	5,684	1,705	195 (17)

3.94%

9.04%

RQ2) Reduced Manual Effort

Object	#	Baseline		MiniPoly		MiniPoly-opt		Rules
		core-js	es-shims	Auto	Man.	Auto	Man.	
Array	24	2,298	8,978	1,332	317			
String	12	1,281	6,885	418	130			
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Total†	89	14,312	32,836	5,714	1,295	5,684	1,705	195 (17)

5.79%

13.28%

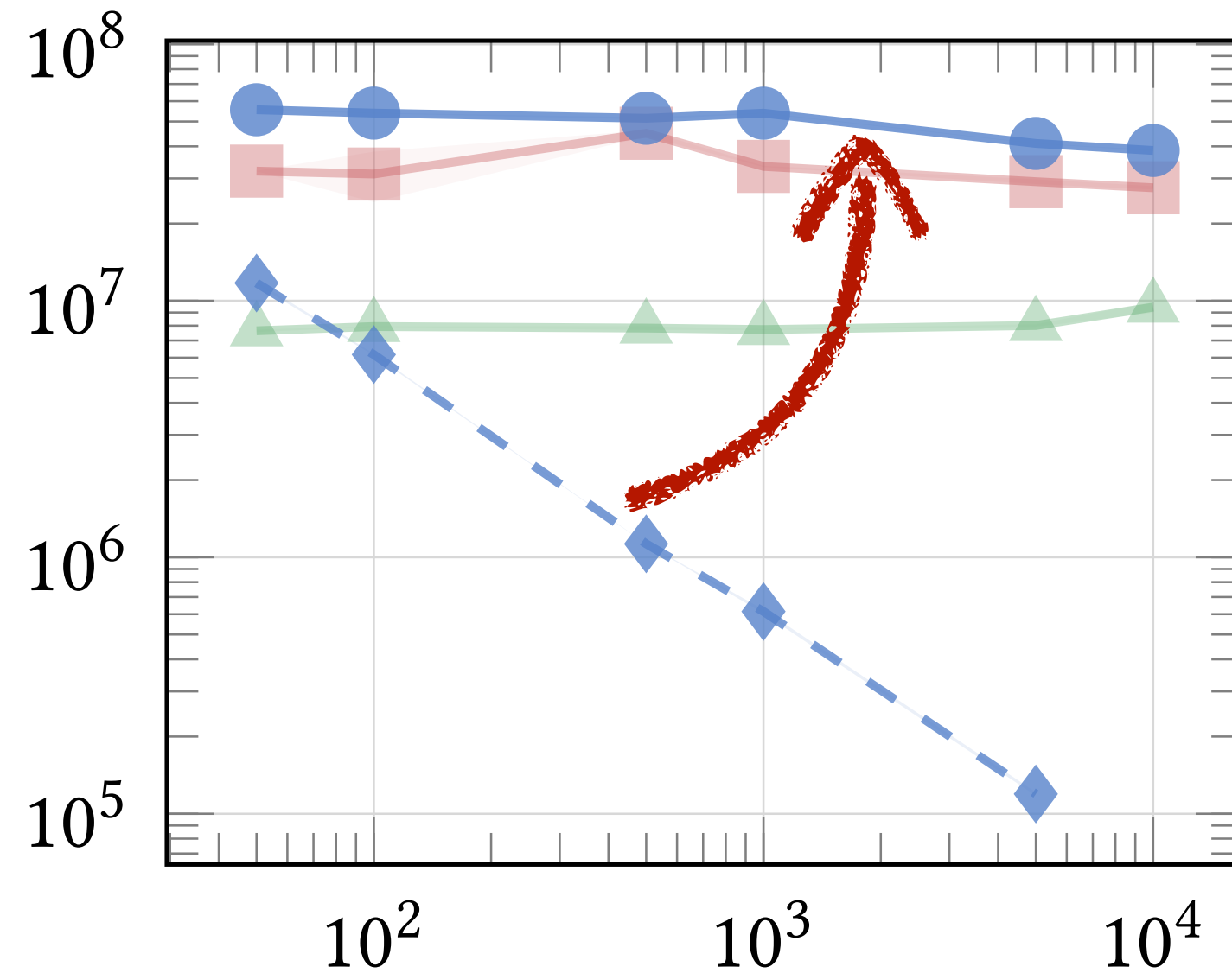
17 rules

RQ3) Speed

- We measured **throughput** (*y-axis* = ops/sec) across **sizes** (*x-axis* = N)

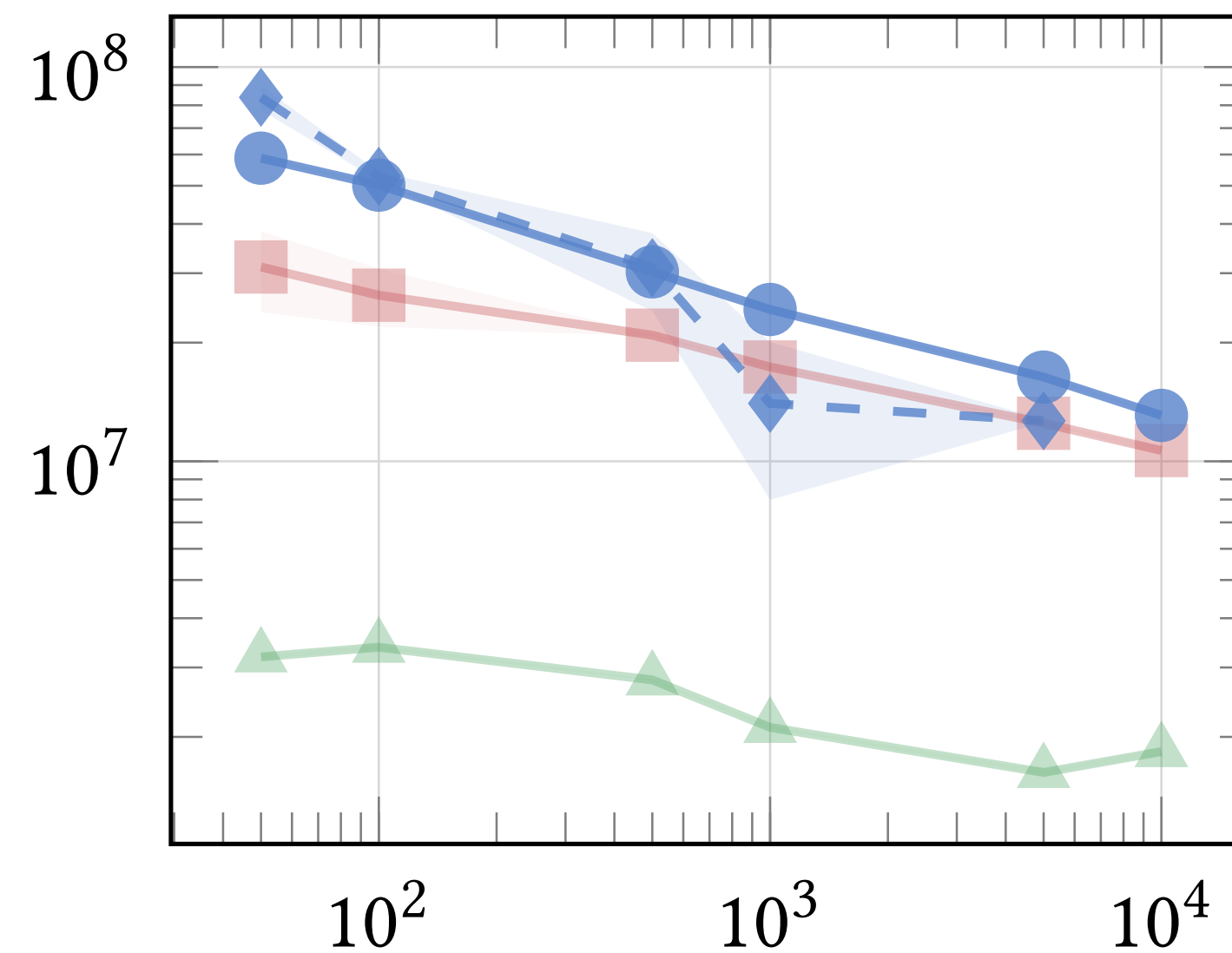
$O(N) \rightarrow O(1)$

Set — Simple



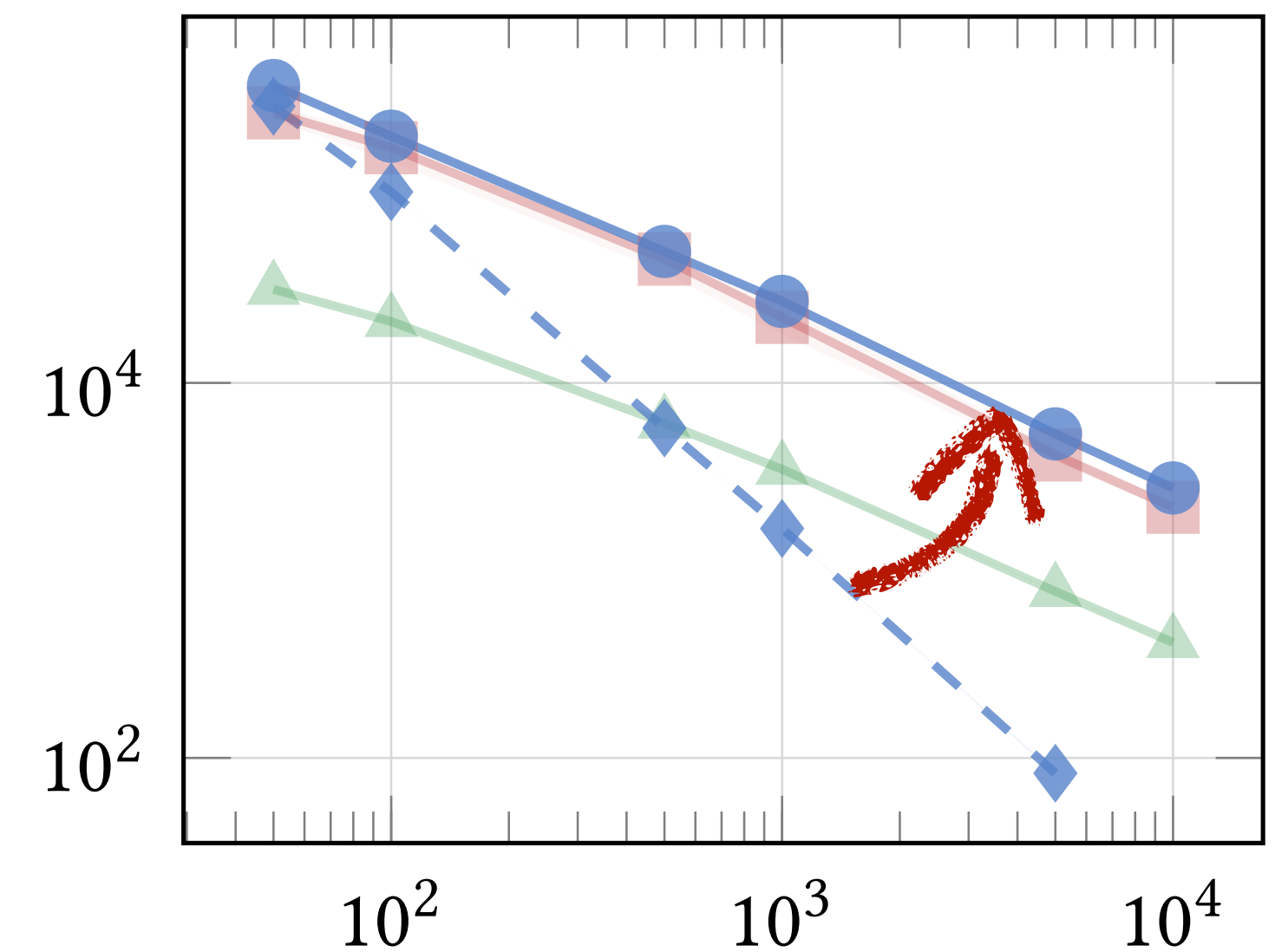
$O(N) \rightarrow O(N)$

Set — Iterate



$O(N^2) \rightarrow O(N)$

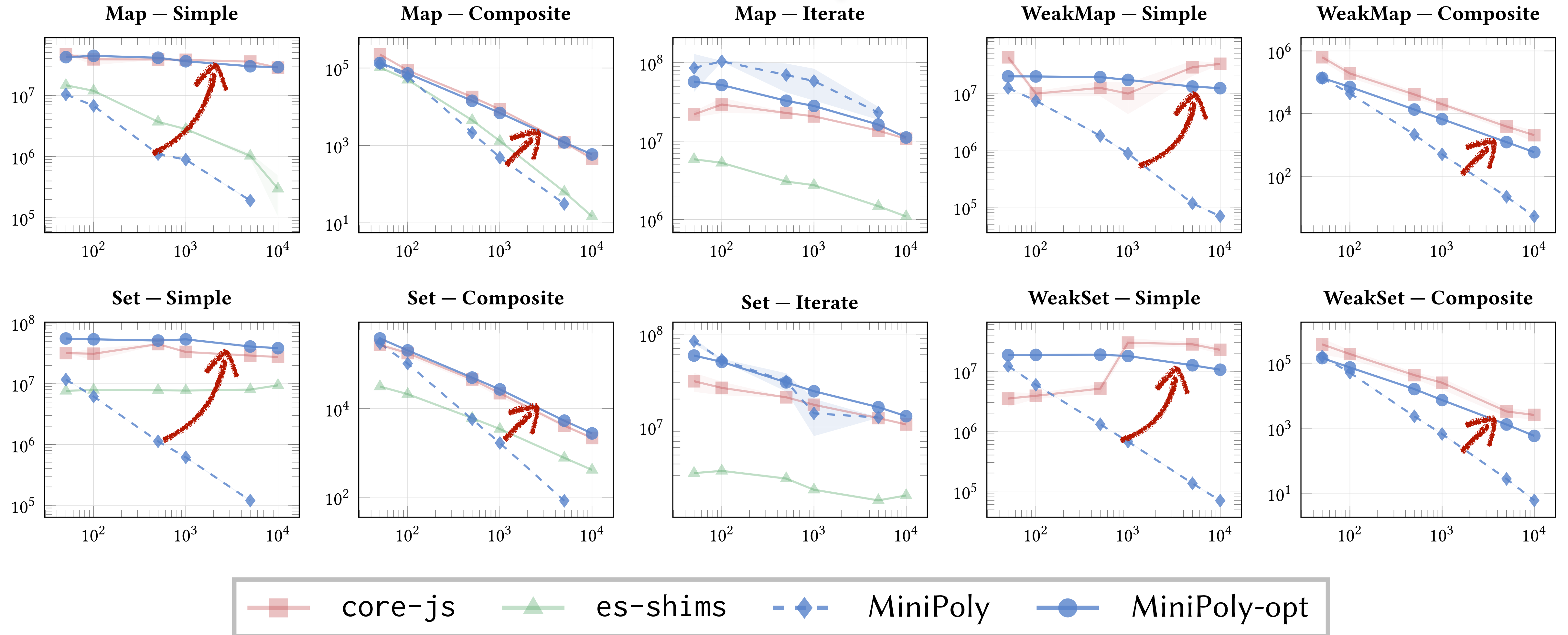
Set — Composite



core-js es-shims MiniPoly MiniPoly-opt

RQ3) Speed

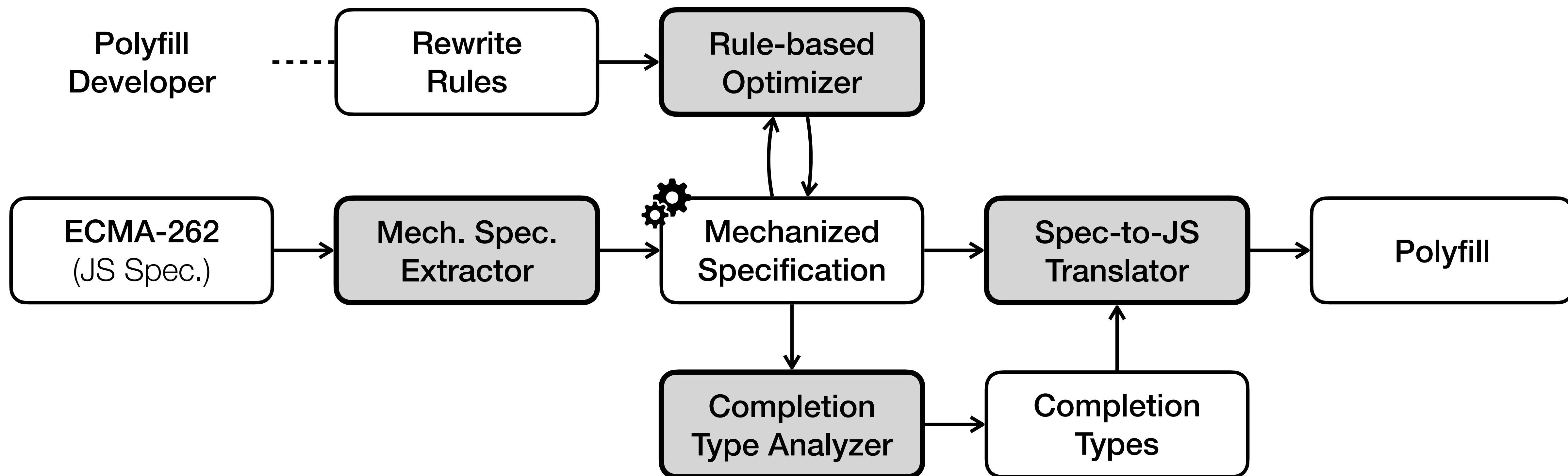
Competitive with baselines in most cases

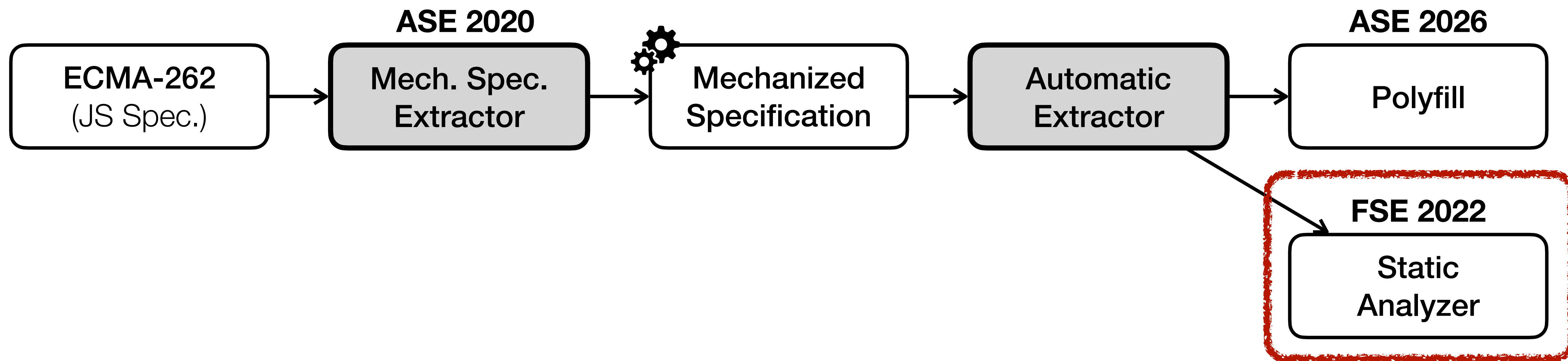


RQ4) Forward Compatibility

- Supports Built-in APIs in **ES2026** and **Stage 2.7** with **100% correctness** and **compatible speed** with manual baselines

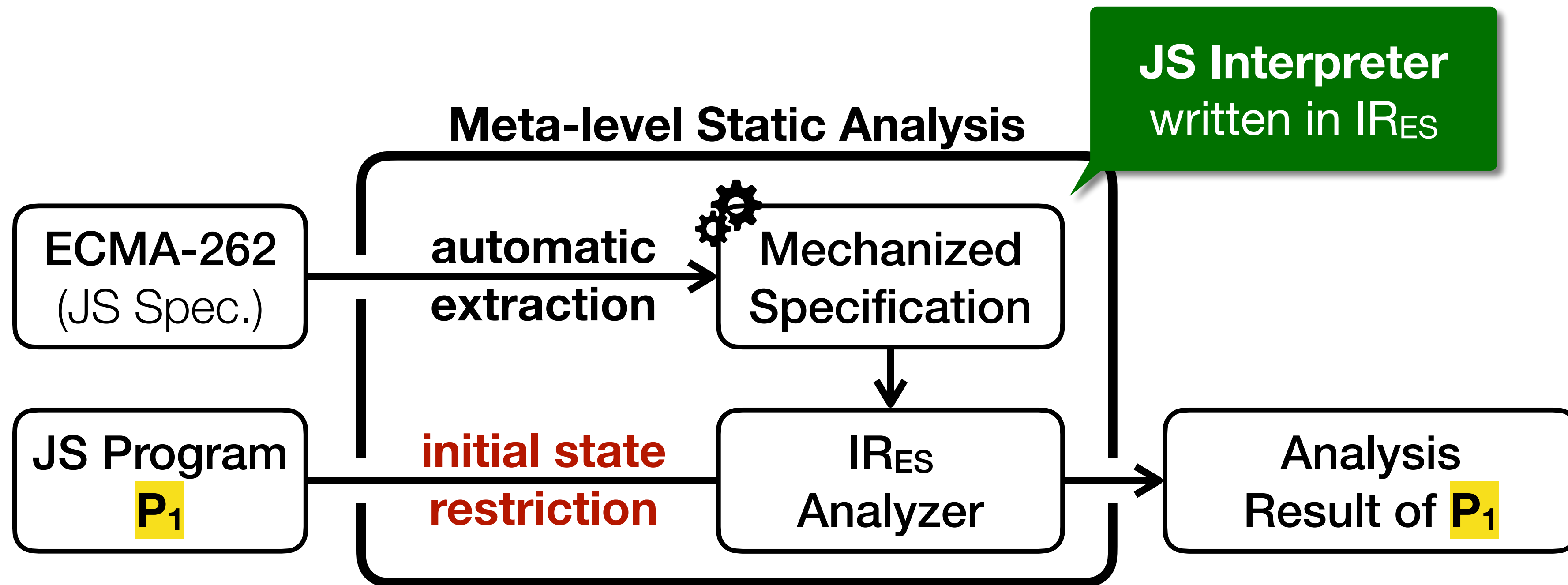
Proposal	Method	Tests	Correctness (Pass %)				Throughput (ops/sec) — N = 5,000			
			core-js	es-shims	MiniPoly	MiniPoly-opt	core-js	es-shims	MiniPoly	MiniPoly-opt
Upsert (ES2026)	Map.prototype									
	.getOrInsert	28	26 (92.9)	24 (85.7)	28 (100.0)	28 (100.0)	13.7M / 2.0M	61.2K	107K	8.2M / 2.5M
	.getOrInsertComputed	37	35 (94.6)	33 (89.2)	37 (100.0)	37 (100.0)	11.7M / 1.8M	40.8K	77.7K	6.5M / 1.9M
	WeakMap.prototype									
	.getOrInsert	34	26 (76.5)	—	34 (100.0)	34 (100.0)	18.4M / 3.1M	—	104K	8.2M / 2.3M
	.getOrInsertComputed	43	32 (74.4)	—	43 (100.0)	43 (100.0)	5.7M / 1.2M	—	73.1K	5.0M / 1.8M
Await Dictionary (Stage 2.7)	Promise									
	.allKeyed	12	—	—	12 (100.0)	12 (100.0)	—	—	—	—
	.allSettledKeyed	12	-23	-8	12 (100.0)	12 (100.0)	—	—	—	—
Total		166	119 (83.8)	57 (87.7)	166 (100.0)	166 (100.0)	11.3M / 1.9M	50.0K	89.3K	6.8M / 2.1M





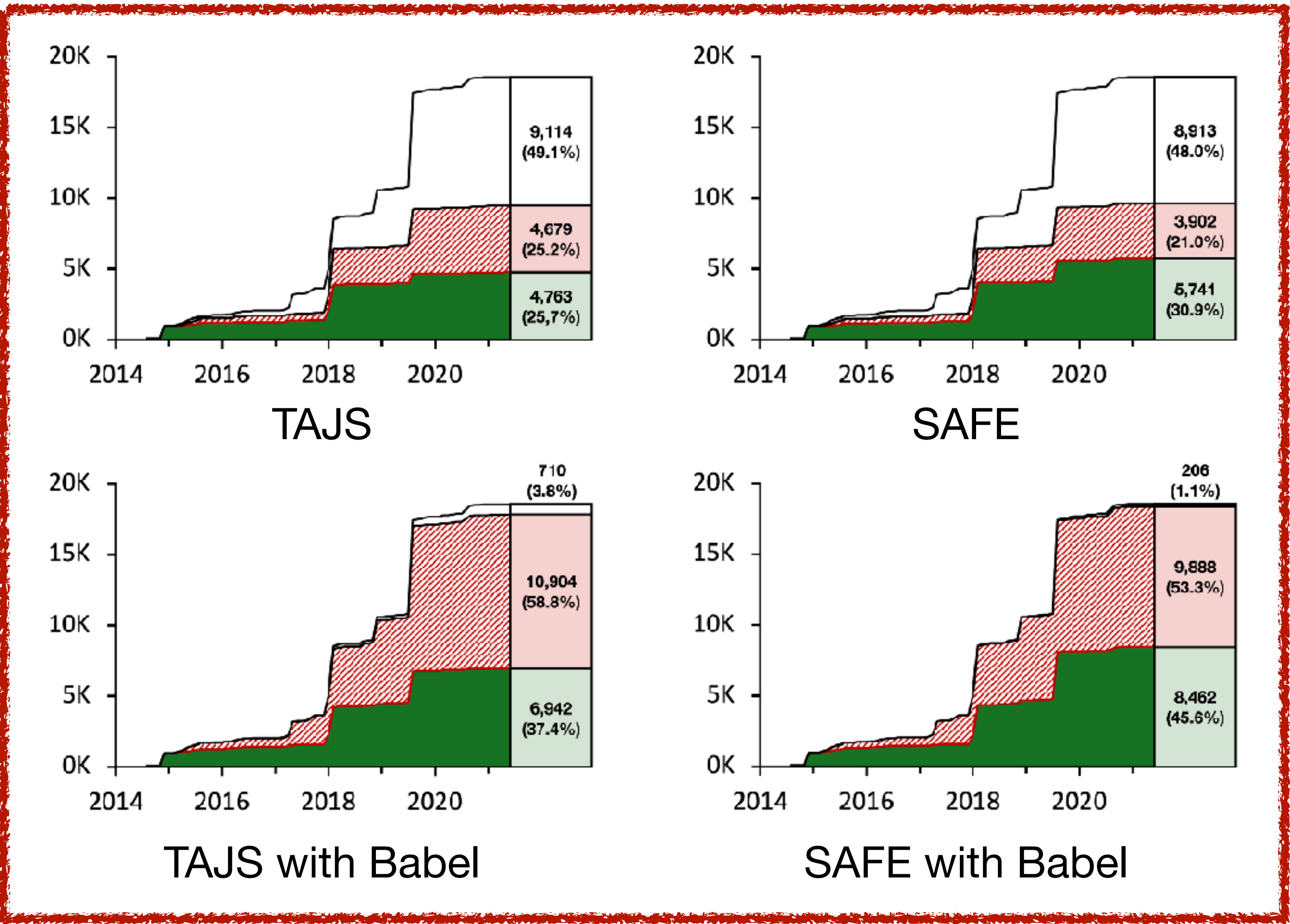
Meta-Level Static Analysis

How to perform **static analysis** on **JavaScript** programs using **language specification**?

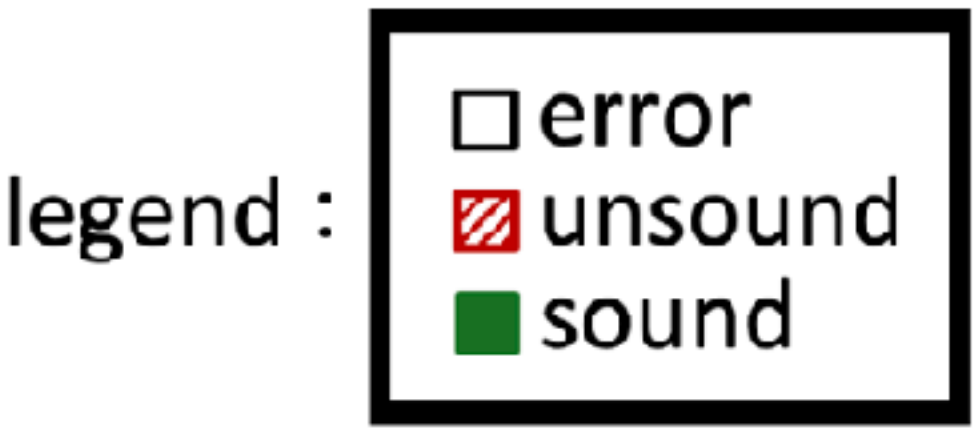
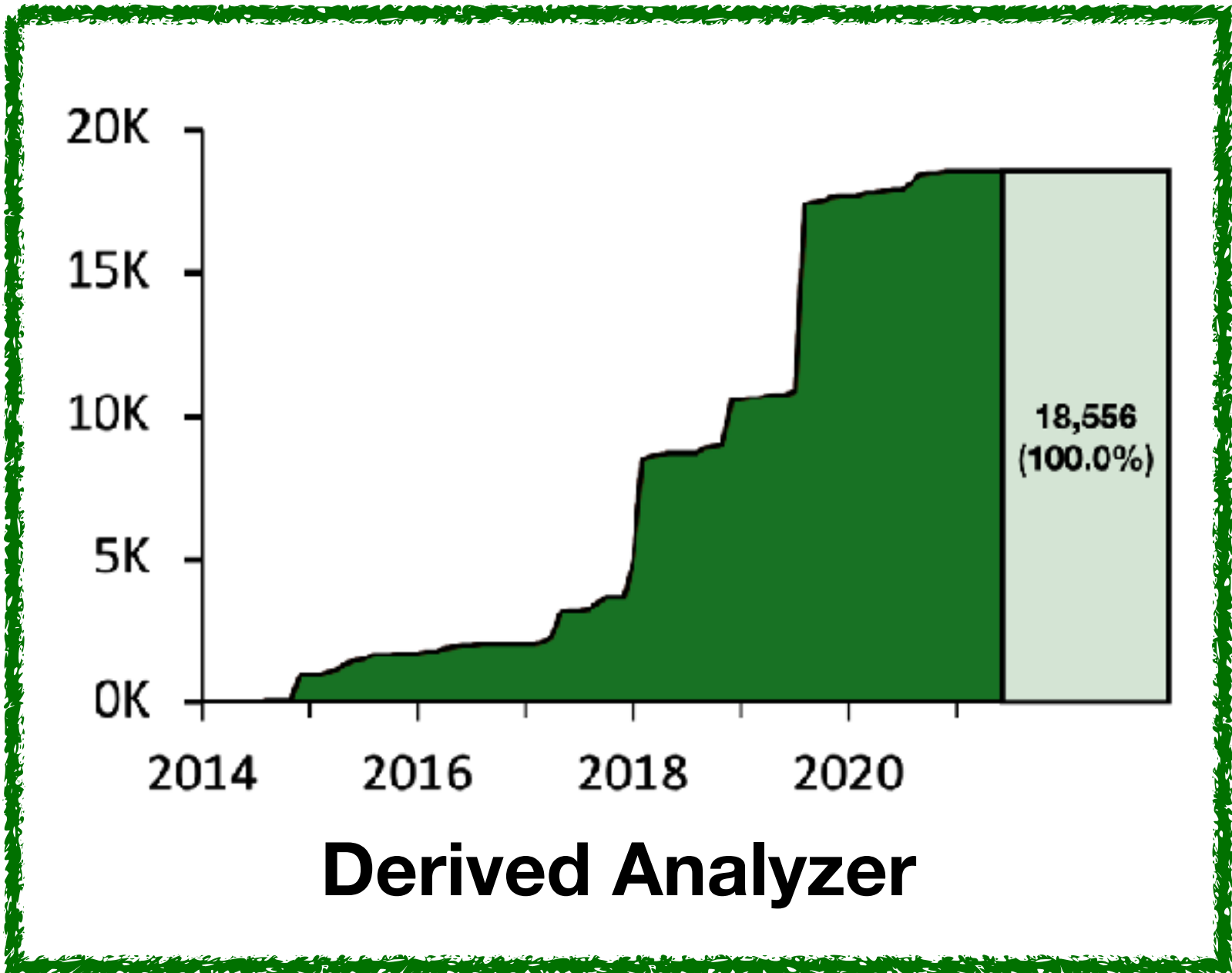


Evaluation

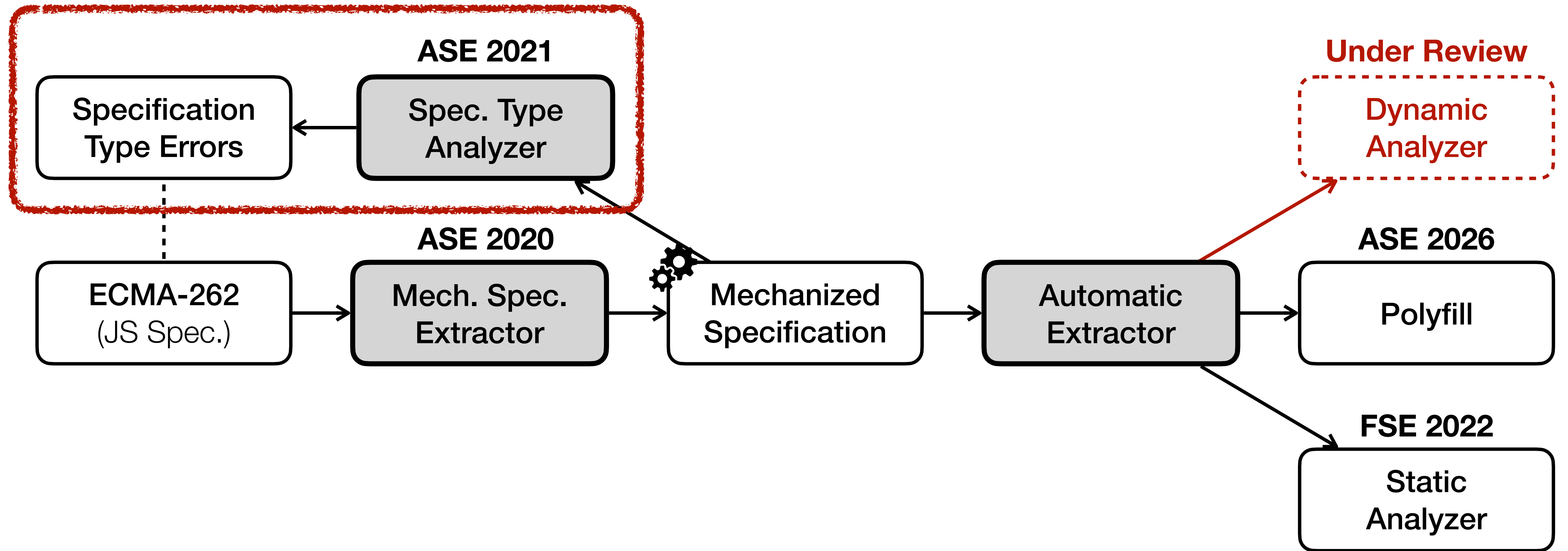
Manual



Derived



x-axis : creation time (year)
y-axis : # tests



Specification Type Analysis

20.3.2.28 Math.round (x)

$n : \text{Number}$

Filter Exception

$x : \text{String} \mid \text{Boolean} \mid \text{Number} \mid \text{Object} \mid \dots$

$\text{Number} \mid \text{Exception}$

1. Let n be $\text{ToNumber}(x)$.
2. If n is an integral Number, return n .
3. If $x < 0.5$ and $x > 0$, return $+0$.
4. If $x < 0$ and $x \geq -0.5$, return -0 .
- ...

Type Error:
'<', '>', and '>='
are numeric operators

$\text{Math.round}(\text{true}) = ???$
 $\text{Math.round}(\text{false}) = ???$

3. If $n < 0.5$ and $n > 0$, return $+0$.
4. If $n < 0$ and $n \geq -0.5$, return -0 .

Fixed

$\text{Math.round}(\text{true}) = 0$
 $\text{Math.round}(\text{false}) = 1$

<https://github.com/tc39/ecma262/tree/575149cfd77aebcf3a129e165bd89e14caafc31c>

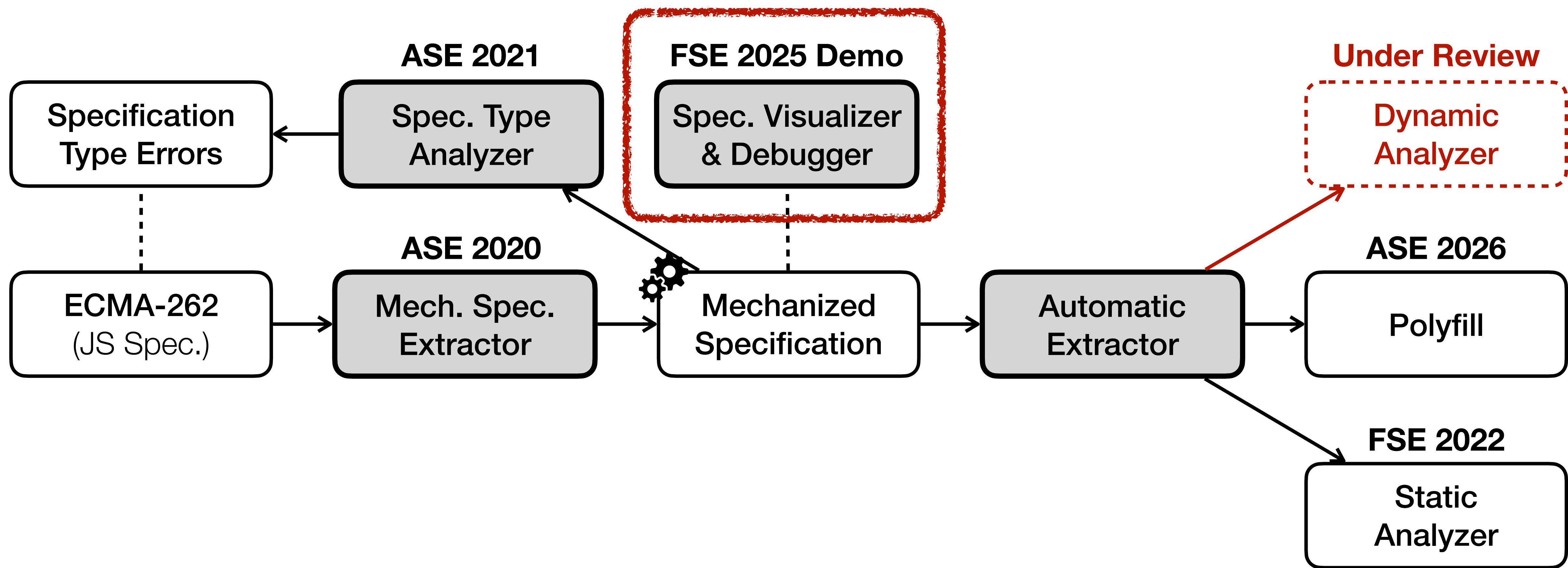
Evaluation

- Type analysis on **864 versions** of ECMA-262 in **3 years**

Checker	Bug Kind	Precision = (# True Bugs) / (# Detected Bugs)					
		no-refine		refine		Δ	
Reference	UnknownVar	62 / 106	17 / 60	63 / 78	17 / 31	+1 / -28	/ -29
	DuplicatedVar		45 / 46		46 / 47		+1 / +1
Arity	MissingParam	93 Errors Detected		4 / 4	4 /	59.2% Precision	
Assertion	Assertion			4 / 31	4 / 3		
Operand	NoNumber			22 / 44	2 /		
	Abrupt			22 / 115	20 / 48		
Total		92 / 279 (33.0%)		93 / 157 (59.2%)		+1 / -122 (+26.3%)	

Name	Feature	#	Checker	Created	Life Span
ES12-1	Switch	3	Reference	2015-09-22	1,996 days
ES12-2	Try	3	Reference	2015-09-22	1,996 days
ES12-3	Arguments	1	Reference	2015-09-22	1,996 days
ES12-4	Array	2	Reference	2015-09-22	1,996 days
ES12-5	Async	1	Reference	2015-09-22	1,996 days
ES12-6	Class	1	Reference	2015-09-22	1,996 days
ES12-7	Branch	1	Reference	2015-09-22	1,996 days
ES12-8	Arguments	2	Operand	2015-12-16	1,910 days

14 New Bugs
In ES2021



▶ RUN

□ QUIT

⏏ SPEC STEP

↺ OVER

↑ OUT

⏏ CONTINUE

↶ SPEC BACK

↶ BACK OVER

↑ BACK OUT

⏮ REWIND

⏏ JS STEP

⏏ AST STEP

↺ OVER

↑ OUT

<> JavaScript Editor ▾

☰ ECMAScript Specification ▾

☰ State

📦 Env

🗑 Heap

⊗ Breaks

☰ Callstack

```
1 const a = new Set
  ([1, 2, 3]);
2 const b = new Set
  ([1, 2]);
3 a.isSupersetOf(b);
```

Context Not Found

Disabled. Start debugger to use.

▶ RUN

□ QUIT

⏏ SPEC STEP

↺ OVER

↑ OUT

⏏ CONTINUE

↶ SPEC BACK

↶ BACK OVER

↑ BACK OUT

⏮ REWIND

⏏ JS STEP

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↺ OVER

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<> JavaScript Editor ▾

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1 const a = new Set
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  ([1, 2]);
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```

☰ ECMAScript Specification ▾

Set.prototype.isSupersetOf(this, ArgumentsList, NewTarget) 🔗

1. Let *O* be the **this** value.
2. Perform ? RequireInternalSlot(*O* , [[SetData]]).
3. Let *otherRec* be ? GetSetRecord(*other*).
4. If SetDataSize(*O* .[[SetData]]) < *otherRec* .[[Size]], return **false**.
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☰ State

📦 Env

🗑 Heap

⛔ Breaks

☰ Callstack

Breakpoints

⏏ USING BREAKPOINTS

search by name

Step

Name

Enable

Remove

No breakpoints. Add Breakpoint by clicking on steps in spec viewer or by searching name

▶ RUN

□ QUIT

⏏ SPEC STEP

↺ OVER

↑ OUT

⏏ CONTINUE

↶ SPEC BACK

↶ BACK OVER

↑ BACK OUT

⏮ REWIND

⏏ JS STEP

⏏ AST STEP

↺ OVER

↑ OUT

<> JavaScript Editor

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≡ State

Env

≡ Heap

⊗ Breaks

≡ Callstack

≡ Specification Environment

- **ArgumentsList** : List
- **NewTarget** : undefined
- **0** : Record[Object]
- **other** : Record[Object]
- **this** : Record[Object]

≡ JavaScript Environment

No environment variables.

▶ RUN

□ QUIT

⏏ SPEC STEP

↺ OVER

↑ OUT

⏏ CONTINUE

↶ SPEC BACK

↶ BACK OVER

↑ BACK OUT

⏮ REWIND

⏏ JS STEP

⏏ AST STEP

↺ OVER

↑ OUT

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State Env Heap Breaks Callstack

Specification Environment

- **ArgumentsList** : List
- **NewTarget** : undefined
- **0** : Record[Object]
- **other** : Record[Object]
- **this** : Record[Object]

JavaScript Environment

No environment variables.

▶ RUN

□ QUIT

⏏ SPEC STEP

↺ OVER

↑ OUT

⏏ CONTINUE

↶ SPEC BACK

↶ BACK OVER

↑ BACK OUT

⏮ REWIND

⏏ JS STEP

⏏ AST STEP

↺ OVER

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≡ State

Env

≡ Heap

⊗ Breaks

≡ Callstack

≡ Specification Environment

- **ArgumentsList** : List
- **NewTarget** : undefined
- **0** : Record[Object]
- **other** : Record[Object]
- **otherRec** : Record[SetRecord]
- **this** : Record[Object]

≡ JavaScript Environment

No environment variables.

▶ RUN

□ QUIT

⏏ SPEC STEP

↺ OVER

↑ OUT

⏏ CONTINUE

↶ SPEC BACK

↶ BACK OVER

↑ BACK OUT

⏮ REWIND

⏏ JS STEP

⏏ AST STEP

↺ OVER

↑ OUT

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≡ State

Env

≡ Heap

⊗ Breaks

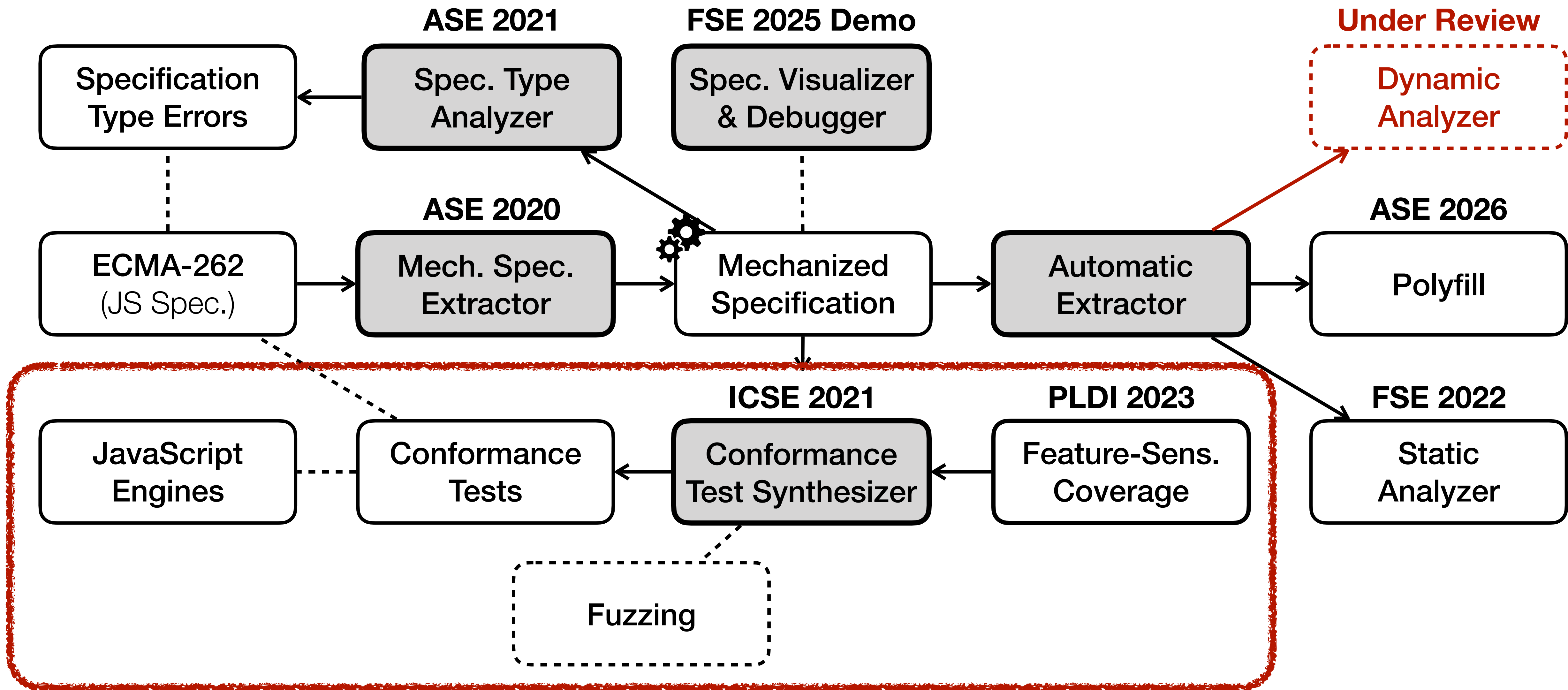
≡ Callstack

≡ Specification Environment

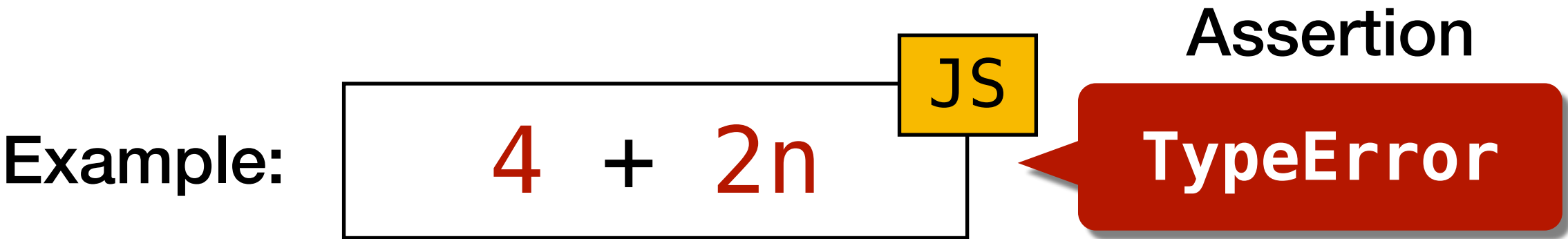
- **ArgumentsList** : List 🔍 Q ✓
- **NewTarget** : undefined
- **0** : Record[Object] 🔍 Q ✓
- **other** : Record[Object] 🔍 Q ✓
- **otherRec** : Record[SetRecord] 🔍 Q ✓
- **Size** : 2
- **Keys** : Record[BuiltinFunctionObject] 🔍 Q ✓
- **Has** : Record[BuiltinFunctionObject] 🔍 Q ✓
- **SetObject** : Record[Object] 🔍 Q ✓
- **this** : Record[Object] 🔍 Q ✓

⊞ JavaScript Environment

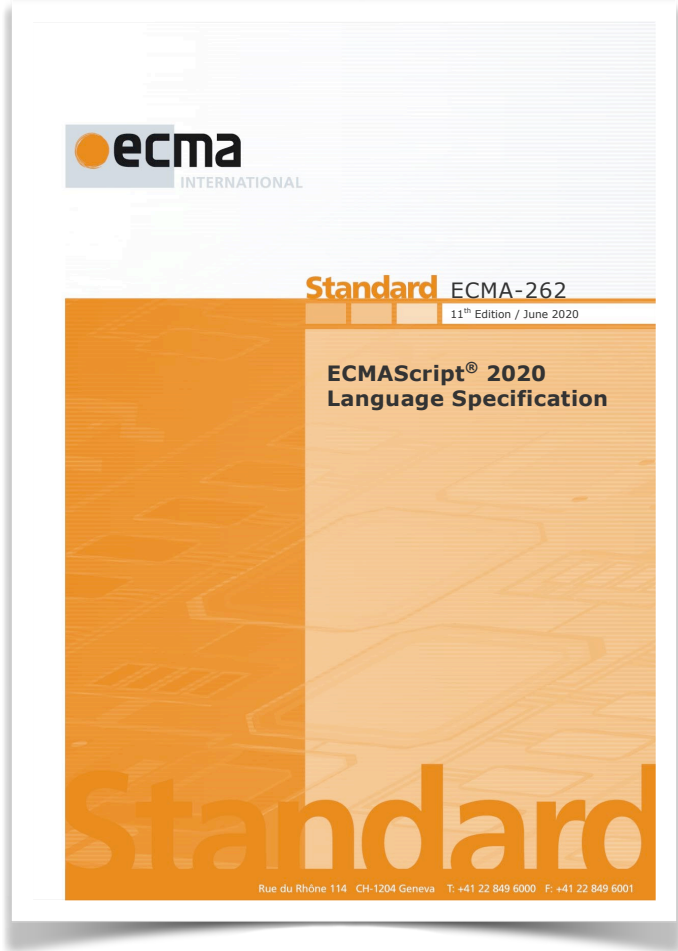
No environment variables.



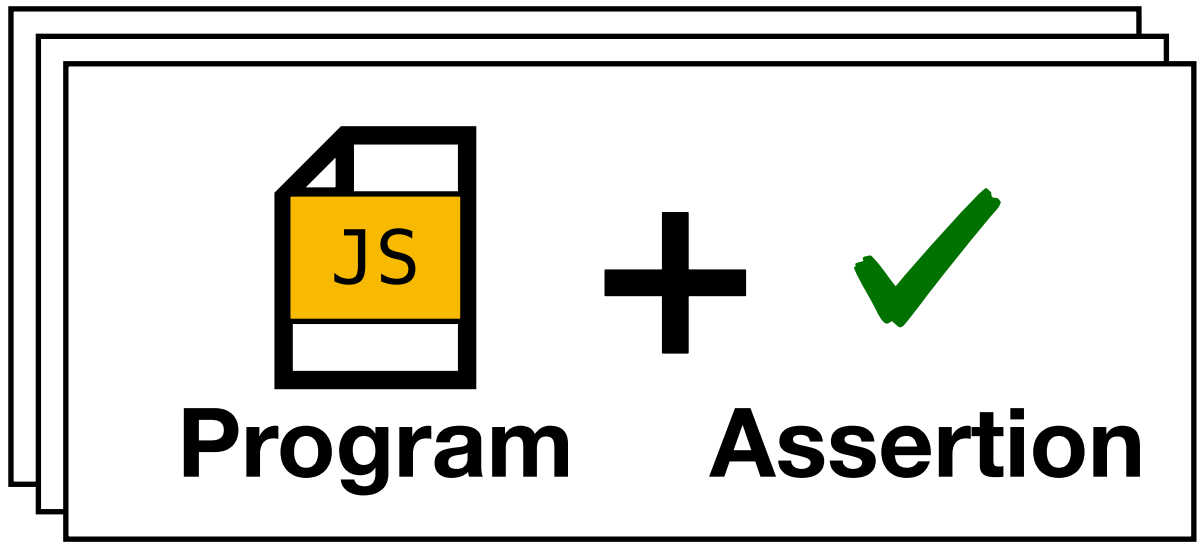
Conformance of JavaScript Engines



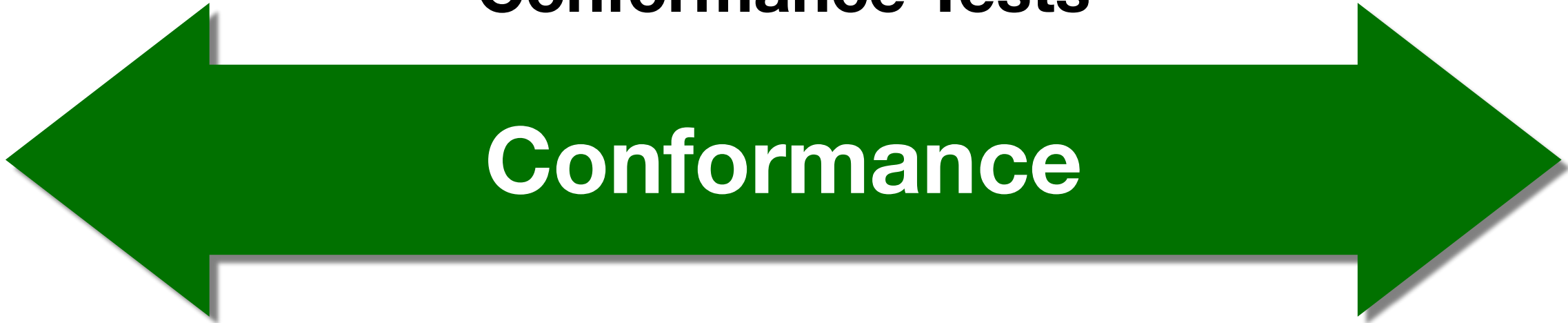
JavaScript
Engines



ECMA-262
(JavaScript Spec.)

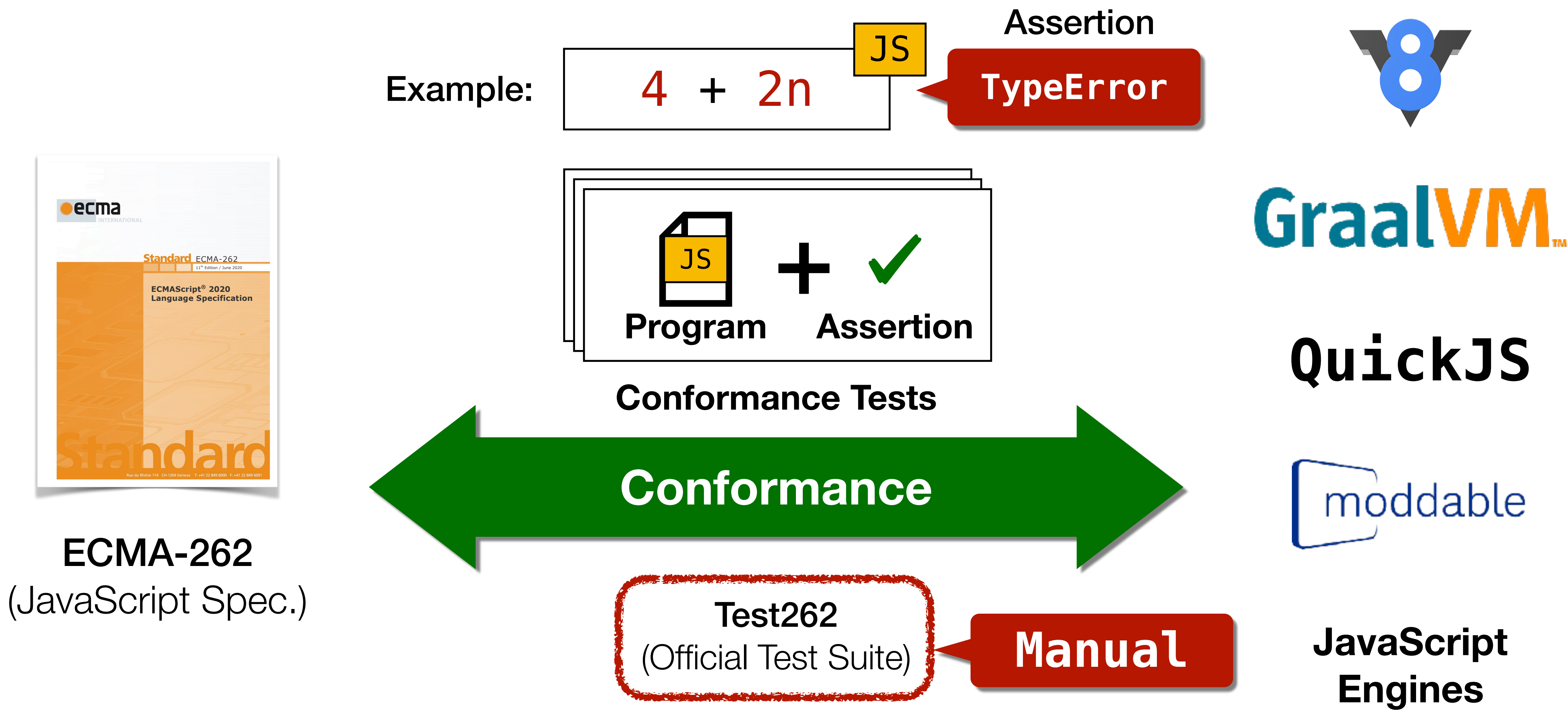


Conformance Tests

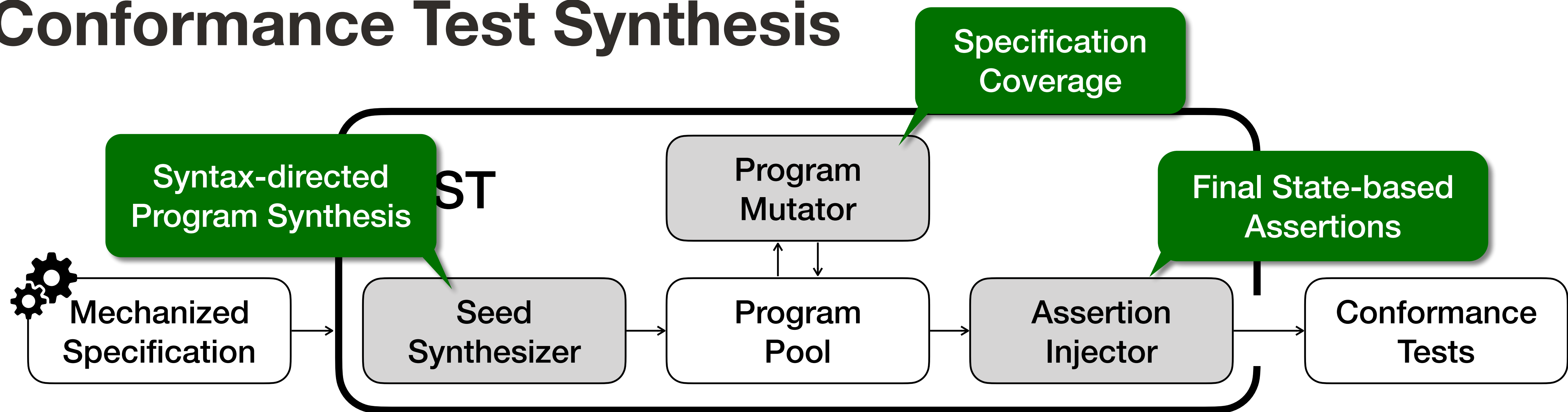


Test262
(Official Test Suite)

Problem - Manual Approach



Conformance Test Synthesis



Program Pool

```
...  
let x = 1 + 2;  
assert(x == 3);  
...  
  
let x = 42;  
assert(x == 42);  
...  
  
let x = ![];  
assert(x == false);  
...
```

Specification Coverage

ApplyStringOrNumericBinaryOperator (*lval*, *opText*, *rval*)

...

5. If **Type**(*lnum*) is not **Type**(*rnum*), throw a **TypeError** exception.

6. If *lnum* is a **BigInt**, then

...

7. Else,

...

3 + 2

JS

1n - 2n

JS

4 * 2n

JS

Feature-Sensitive Coverage

ApplyStringOrNumericBinaryOperator (*lval*, *opText*, *rval*)

...

5. If `Type(lnum)` is not `Type(rnum)`, throw a **TypeError** exception.

6. If *lnum* is a `BigInt`, then

...

7. Else,

...

Language
Features

Addition

3 + 2

JS

1n + 2n

JS

17 + 2n

JS

Subtraction

4 - 7

JS

7n - 2n

JS

1n - 23

JS

Multiplication

2 * 9

JS

9n * 3n

JS

3n * 45

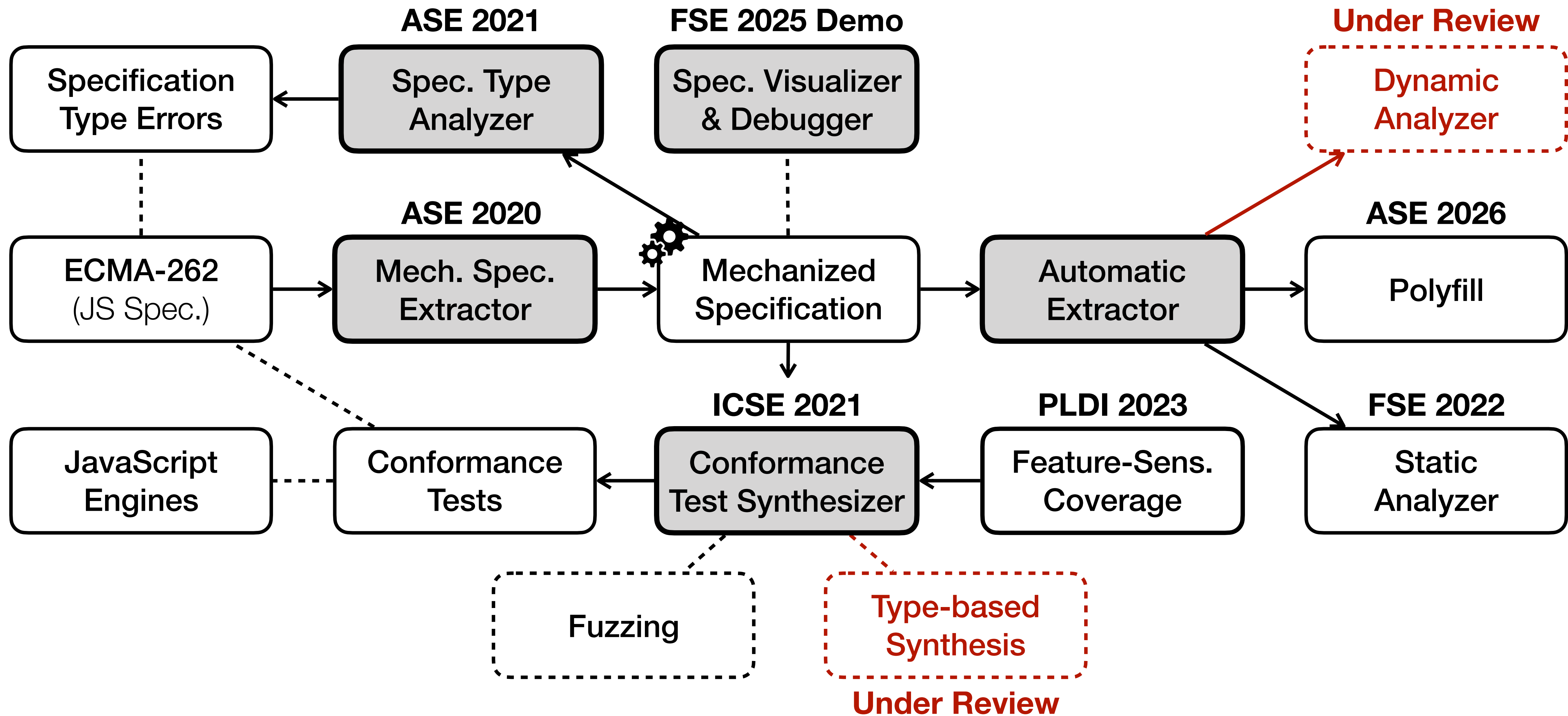
JS

Specification Coverage

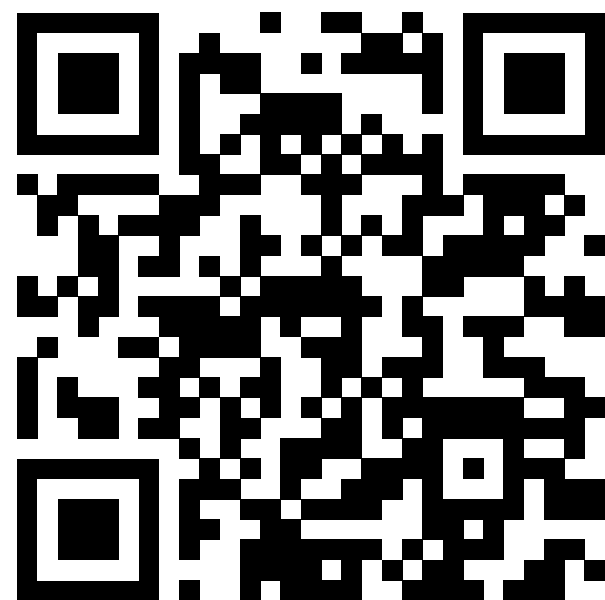
Kind	Name	Version	Release	# Detected Unique Bugs		
				# New	# Confirmed	# Reported
Engine	V8	v10.8.121	2022.10.06	3	3	4
	JSC	v615.1.10	2022.10.26	26	26	26
	GraalJS	v22.2.0	2022.07.26	11	11	11
	SpiderMonkey	v107.0b4	2022.10.24	2	4	4
	Total			42	44	45
Transpiler	Babel	v7.19.1	2022.09.15	37	37	39
	SWC	v1.3.10	2022.10.21	37	37	47
	Terser	v5.15.1	2022.10.05	19	19	19
	Obfuscator	v4.0.0	2022.02.15	1	2	7
	Total			94	95	112
Total				136	139	157

42 Bugs
In Engines

94 Bugs
In Transpilers



Official tool used in CI system of
ECMA-262 and Test262



<https://github.com/es-meta/esmeta>

